

- Q1** What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?
 (A) 10 / 5 (B) 3 / 5
 (C) 2 / 5 (D) 1 / 5
- Q2** Which of the following is not a palindromic subsequence of the string "ababcbabba"?
 (A) abcba (B) abba
 (C) abbbba (D) adba
- Q3** Which of the following is TRUE?
 (A) The cost of searching an AVL tree is $\theta(\log n)$ but that of binary search is $O(n)$
 (B) The cost of searching an AVL tree is $\theta(\log n)$ but that of complete binary tree is $\theta(n \log n)$
 (C) The cost of searching a binary tree is $O(\log n)$ but that of AVL tree is $\theta(n)$
 (D) The cost of searching an AVL tree is $\theta(n \log n)$ but that of a binary search tree is $O(n)$
- Q4** What is the result of evaluating the postfix expression " 43*25*+8- "?
 (A) 8 (B) 14
 (C) 10 (D) 5
- Q5** Which of the following circuits is used to store one bit of data?
 (A) Encoder (B) Decoder
 (C) Flip-Flop (D) Register
- Q6** Identify the code sequence:
 1010
 1011
 1001
 1000
 (A) BCD (B) Excess-3
 (C) Gray (D) Excess-3 Gray
- Q7** The microoperation which divides a signed binary number by 2 is:
 (A) Circular shift
 (B) Logical shift
 (C) Arithmetic shift right
 (D) Arithmetic shift left
- Q8** A program that is used by other routines to accomplish a particular task, is called:
 (A) Micro program
 (B) Micro operation
 (C) Routine
 (D) Subroutine
- Q9** Consider a triangle PQR with coordinates as P(0,0), Q(2,2) and R(10,4). If this triangle is to be magnified to four times its size while keeping R(10,4) fixed, then the coordinates of the magnified triangle are :
 (A) (-20,-12), Q(-20,-4) and R(10,4)
 (B) (-30,-12), Q(-22,-4) and R(10,4)
 (C) (-25,-10), Q(22,-4) and R(10,4)
 (D) (30,-12), Q(-22,4) and R(10,4)
- Q10** Which of the following statements is TRUE?
 (A) Virtual functions do not implement polymorphism
 (B) Virtual functions do not permit calling of derived class functions using a base class pointer
 (C) We can never build an object from a class containing a pure virtual function
 (D) Pure virtual function can never have a body
- Q11** What is the output of the following program?

```
#include<stdio.h>
int main()
{ int i=3;
  while (i--)
  { int i=10;
    i--;
    printf("%d", i );
  }
  printf("%d", i );
}
```

 (A) 990 (B) 9990
 (C) 999-1 (D) 99-1



- Q12** The Hue of a color is related to its:
 (A) Luminance (B) Saturation
 (C) Incandescence (D) Wavelength
- Q13** "CREATE TABLE T" in SQL is an example of:
 (A) Normalization (B) DML
 (C) DDL (D) Primary key
- Q14** An Address in main memory is called:
 (A) Virtual address
 (B) Memory address
 (C) Logical address
 (D) Physical address
- Q15** A system bus in which each data item is transferred during a time slice known in advance to both unit source and destination is called :
 (A) MIMD
 (B) DMA
 (C) asynchronous bus
 (D) synchronous bus
- Q16** What is the output of the following program?

```
# include <stdio.h>
# define SQR(x)(x* x)
int main ( )
{int a, b=3;
a=SQR(b+2);
printf("%d", a);
return 0 ;
}
```

 (A) 25 (B) 11
 (C) Garbage value (D) 24
- Q17** The head of a moving head disk with 200 tracks, numbered 0 to 199, has just finished a request at track 125, and currently serving a request at track 143. The queue of requests is given in the FIFO order as 86,147,91,177,94,150,102,175,130. What will be the total number of head movements required to satisfy these requests for SCAN algorithm?
 (A) 259 cylinders (B) 169 cylinders
 (C) 154 cylinders (D) 264 cylinders

Q18

Given as 4GB (~approx 4.3×10^9 bytes) of virtual space and a typical page size of 4 KB and each page table entry is 5 bytes. How many virtual pages would this imply? What is the size of the whole page table?

- (A) 1075000 and 5375 KB
 (B) 215000 and 40960 bytes
 (C) 10750 and 10240 bytes
 (D) 43000 and 1024 bytes

- Q19** In Linux, where is the user password stored?
 (A) / etc/password
 (B) / root/password
 (C) /etc/passwd
 (D) / root/passwd

- Q20** Hash functions are used to produce the message digests which are then encrypted with a private key to get:
 (A) Public key
 (B) Digital signature
 (C) Cipher text
 (D) Data Encryption Standard

- Q21** Let $L = \{ab, aa, baa\}$. Which of the following strings are not in L^* ?
 (A) abaabaaabaa (B) aaaabaaaa
 (C) baaaaabaaaab (D) baaaaabaa

- Q22** Let $A = \{a, b\}$ and $L = A^*$. Let $X = \{a^n b^n, n > 0\}$. The languages $L \cup X$ and X are respectively :
 (A) Not regular, Regular
 (B) Regular, Regular
 (C) Regular, Not regular
 (D) Not Regular, Not Regular

- Q23** Which of the following symbol table implementations is best suited if access time is to be minimum?
 (A) Linear list
 (B) Search tree
 (C) Hash Table
 (D) Self organisation list

- Q24** Consider a Grammar $E \rightarrow E+n \mid Exn \mid n$ for a sentence $n+n \times n$ the handles in the right-sentential form of the reduction are?



- (A) n , $E+n$ and $E+n \times n$
 (B) n , $E+n$ and $E \times n$
 (C) n , $n+n$ and $n+n \times n$
 (D) n , $E+n$ and $E \times n$
- Q25** Which data structure is typically used to implement a hash table?
 (A) Linked list (B) Array
 (C) Binary Tree (D) Stack
- Q26** Which collision resolution technique involves maintaining a linked list of collided keys?
 (A) Linear probing
 (B) Quadratic probing
 (C) Chaining
 (D) Double hashing
- Q27** Consider the following functions :
 $f(n)=3n\sqrt{n}$
 $g(n)=2^{\sqrt{n}\log_2 n}$
 $h(n)=n!$
 Which of the following is true?
 (A) $h(n)$ is $O(f(n))$
 (B) $h(n)$ is $O(g(n))$
 (C) $g(n)$ is not $O(f(n))$
 (D) $f(n)$ is $O(g(n))$
- Q28** 2-3-4 trees are B - trees of order 4. They are isometric of trees.
 (A) AVL (B) AA
 (C) 2-3 (D) Red-Black
- Q29** The average time required to search a storage location in memory and obtain its contents is called :
 (A) Access time (B) Latency time
 (C) Response time (D) Reading time
- Q30** One of the purposes of using intermediate code in compilers is to :
 (A) make parsing and semantic analysis simpler
 (B) improve error recovery and error reporting
 (C) increase the chances of reusing the machine independent code optimizer in other compilers
 (D) improve the register allocation
- Q31** The sum of minimum and maximum number of final states for a Deterministic Finite Automata (DFA) having 'P' state is equal to :
 (A) P (B) P-1
 (C) P+1 (D) P+2
- Q32** Test suite is consist of :
 (A) Set of defect cases
 (B) Set of boundary cases
 (C) Set of test cases
 (D) Set of nest cases
- Q33** A multiplexes combines four 100 Kbps channels using a time slot of 2 bits. What is the bit rate?
 (A) 100 Kbps (B) 200 Kbps
 (C) 400 Kbps (D) 1000 Kbps
- Q34** Which of the following is not a field in TCP header?
 (A) Sequence Number
 (B) Checksum
 (C) Fragmentation offset
 (D) Window size
- Q35** In "bit stuffing", each frame begins and ends with a bit pattern in hexadecimal?
 (A) 0x8C (B) 0x6F
 (C) 0xFF (D) 0x7E
- Q36** Level - 0 DFD is also called as:
 (A) Use case Diagram
 (B) Sequence Diagram
 (C) Context Diagram
 (D) Prototype Diagram
- Q37** What is the generic structure of a Multi-Agent System (MAS)?
 (A) Single-agent with multiple objectives
 (B) Multiagents with single objectives
 (C) Multiagents with diverse objectives and communication abilities
 (D) Multiagent with two objectives
- Q38** A point of a fuzzy set A is a point $x \in X$ at which $\mu_A(x) = 0.5$
 (A) Core (B) Support
 (C) Crossover (D) α - cut



Q39 In a genetic algorithm optimization problem, the fitness function is defined as $f(x) = x^2 - 4x + 4$. Given a population of four individuals with values of $x: \{1.5, 2.0, 3.0, 4.5\}$

What is the fitness value of the individual that will be selected as the parent for reproduction in one generation?

- (A) 2.25 (B) 6.0
(C) 0.0 (D) 6.25

Q40 In a feed-forward neural network with the following specifications :

The input layer has 4 neurons, hidden layer has 3 neurons, and output layer has 2 neurons using the sigmoid activation function for given input values $[0.5, 0.8, 0.2, 0.6]$ as well as the initial weights for the connections.

W1: $[0.1, 0.3, 0.5, 0.2]$ W2: $[0.2, 0.4, 0.6, 0.2]$ W3: $[0.3, 0.5, 0.7, 0.2]$ Input layer to hidden layer weights

W4 : $[0.4, 0.1, 0.3]$ W5 : $[0.5, 0.2, 0.4]$ Hidden layer to output layer weights

What is the output of the output layer when the given input values are passed through neural network? Round the answer to two decimal places:

- (A) $[0.62, 0.68]$ (B) $[0.72, 0.78]$
(C) $[0.82, 0.88]$ (D) $[0.92, 0.98]$

Q41 If universe of discourse are all real numbers, then which of the following are true?

- (A) $\exists x \forall y (x+y=y)$
(B) $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x-y > 0))$
(C) $\exists x \exists y (((x \leq 0) \wedge (y \leq 0)) \wedge (x-y > 0))$
(D) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$

Choose the correct answer from the options given below :

- (A) (A) and (B) Only
(B) (A), (C) and (D) Only
(C) (A), (B) and (D) Only
(D) (A), (B), (C) and (D) Only

Q42 If the universe of discourse is a set of integers, then which of the followings are TRUE ?

- (A) $\forall n \exists m (n^2 < m)$
(B) $\exists n \forall m (n < m^2)$

- (C) $\exists n \forall m (nm = m)$
(D) $\exists n \exists m (n^2 + m^2 = 6)$
(E) $\exists n \exists m (n+m=4 \wedge n-m=1)$

Choose the correct answer from the options given below :

- (A) (A), (B) and (C) Only
(B) (B) and (C) Only
(C) (C), (D) and (E) Only
(D) (C) and (E) Only

Q43 If $N^2 = N \times N$, N is set of natural numbers and R is relation on N^2 , s.t. $R \subset N^2 \times N^2$ i.e. $\langle x, y \rangle R \langle u, v \rangle \Leftrightarrow xv = yu$, then which of the following are TRUE?

- (A) Reflexive
(B) Symmetric
(C) Transitive
(D) Asymmetric

Choose the correct answer from the options given below :

- (A) (A) and (B) Only
(B) (B) and (C) Only
(C) (A), (C) and (D) Only
(D) (A), (B) and (C) Only

Q44 The interface(s) that provide(s) I/O transfer of data directly to and from the memory unit peripheral is/are termed as:

- (A) DMA (Direct Memory Access)
(B) IOP (Input-Output Processor)
(C) Serial Interface
(D) Parallel Interface

Choose the correct answer from the options given below :

- (A) (A) Only
(B) (B) Only
(C) (A) and (B) Only
(D) (C) and (D) Only

Q45 Which of the following(s) are main memory?

- (A) Virtual memory
(B) Cache memory
(C) RAM
(D) SSD

Choose the correct answer from the options given below :

- (A) (A) and (C) Only



- (B) (B) and (C) Only
 (C) (C) and (D) Only
 (D) (A), (B) and (C) Only

Q46 Which of the following are tautology?

- (A) $(P \rightarrow (P \rightarrow Q)) \rightarrow (P \rightarrow Q)$
 (B) $((P \rightarrow Q) \rightarrow Q) \rightarrow (P \vee Q)$
 (C) $((P \vee \neg P) \rightarrow Q) \rightarrow ((P \vee \neg P) \rightarrow R)$
 (D) $(Q \rightarrow (P \wedge \neg P)) \rightarrow (R \rightarrow (P \wedge \neg P))$

Choose the correct answer from the options given below :

- (A) (A) Only
 (B) (B) Only
 (C) (A) and (B) Only
 (D) (C) and (D) Only

Q47 Consider the three points $P_1(1,2,0)$, $P_2(3,6,20)$ and $P_3(2,4,6)$ and a view point $C(0,0,-10)$. Choose the correct options.

- (A) P_1 obscure P_2 , if viewed from C.
 (B) P_2 obscure P_1 , if viewed from C.
 (C) P_3 does not obscure P_1 if viewed from C.
 (D) P_2 does not obscure P_3 if viewed from C.

Choose the correct answer from the options given below :

- (A) (A), (B) and (C) Only
 (B) (A), (C) and (D) Only
 (C) (B), (C) and (D) Only
 (D) (A), (B) and (D) Only

Q48 Which of the statements are CORRECT?

- (A) Constructors are invoked automatically when the objects are created.
 (B) Constructors do not have return types, not even void, and therefore they cannot return values.
 (C) Constructors cannot be inherited though a derived class can call the base class constructors.
 (D) Constructors can be declared as virtual.

Choose the correct answer from the options given below :

- (A) (A), (B) and (D) Only
 (B) (A), (B) and (C) Only
 (C) (B), (C) and (D) Only
 (D) (A), (C) and (D) Only

Q49 Which one of the following statements is CORRECT?

- (A) Granularity is the size of data items in a database.
 (B) Two operations in a schedule are said to be in conflict if they belong to the same transaction.
 (C) Two schedulers are said to be conflict equivalent if the order of any two conflicting operations is the same in both schedules.
 (D) Write operations that are performed without performing the write operation are known as Blind Writes.

Choose the correct answer from the options given below :

- (A) (A) and (B) Only
 (B) (A), (B) and (C) Only
 (C) (A), (B) and (D) Only
 (D) (B) and (C) Only

Q50 Which of the following is/are NOT CORRECT statements?

- (A) The first record in each block of the data file is known as the actor record.
 (B) Dense index has index entries for every search key value in the data file.
 (C) Searching is harder in the B+ tree than in the B - tree as all external nodes are linked to each other.
 (D) In extendible hashing the size of the directory is just an array of $2d-1$, where d is global depth.

Choose the correct answer from the options given below :

- (A) (A), (B) and (C) Only
 (B) (A), (C) and (D) Only
 (C) (A), (B) and (D) Only
 (D) (A), (B), (C) and (D) Only

Q51 Which of the following statements is/are NOT CORRECT about NUMA ?

- (A) LOAD and STORE instructions are used to access remote memory.
 (B) There is a single address space visible to all CPUs.
 (C) Access to local memory is slower than access to remote memory.



(D) When the access time to remote memory is hidden, the system is called NC - NUMA.

(E) In CC - NUMA, Coherent caches are present.

Choose the correct answer from the options given below :

(A) (A) and (C) Only

(B) (B) and (D) Only

(C) (A) and (E) Only

(D) (C) and (D) Only

Q52 Indexed/grouped allocation is useful as :

(A) It supports both sequential and direct access.

(B) Entire block is available for data.

(C) It does not require lots of space for keeping pointers.

(D) No external fragmentation.

Choose the correct answer from the options given below :

(A) (A) Only

(B) (B) and (C) Only

(C) (B) Only

(D) (A), (B) and (D) Only

Q53 Which of the following statements are CORRECT ?

(A) A process always check state of currently executing process to enter critical schema.

(B) Spin locks use busy waiting.

(C) Periodically testing a variable until some value appears is known as busy waiting.

(D) Critical region is a part of the program, where shared memory is kept.

(E) Printer daemon, continuously checks to see if there are any file to be printed.

Choose the correct answer from the options given below :

(A) (A) and (B) Only

(B) (B) and (C) Only

(C) (B) and (D) Only

(D) (B) and (E) Only

Q54 Three address codes can be represented in special structures known as :

(A) Quadruples

(B) Triples

(C) Patterns

(D) Indirect Triples

Choose the correct answer from the options given below :

(A) (A) and (B) Only

(B) (A), (B) and (D) Only

(C) (B) and (C) Only

(D) (B), (C) and (D) Only

Q55 Which of the statement is/are CORRECT ?

(A) Moore and Mealy machines are finite state machines with output capabilities.

(B) Any given Moore machine has an equivalent Mealy machine.

(C) Any given Mealy machine has an equivalent Moore machine.

(D) Moore machine is not a finite state machine.

Choose the correct answer from the options given below :

(A) (A) and (B) Only

(B) (A), (B) and (C) Only

(C) (B) and (D) Only

(D) (A), (B) and (D) Only

Q56 The statement $P(x): "x=x^2"$. If the universe of disclosure consists of integers, what are the following have truth values:

(A) $P(0)$

(B) $P(1)$

(C) $P(2)$

(D) $\exists x P(x)$

(E) $\forall x P(x)$

Choose the correct answer from the options given below :

(A) (A), (B) and (E) Only

(B) (A), (B) and (C) Only

(C) (A), (B) and (D) Only

(D) (B), (C) and (D) Only

Q57 Which of the following statements are truth statements if universe of disclosure is set of integers:

(A) $\forall n (n^2 \geq 0)$

(B) $\exists n (n^2 = 2)$

(C) $\forall n (n^2 \geq n)$

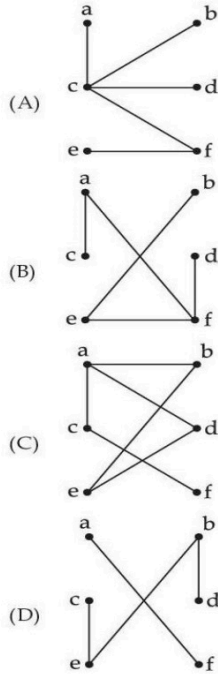
(D) $\exists n (n^2 < 0)$



Choose the correct answer from the options given below :

- (A) (A) and (B) Only
- (B) (B) and (C) Only
- (C) (C) and (D) Only
- (D) (A) and (C) Only

Q58 Which of the following graphs are trees ?



Choose the **correct** answer from the options given below :

- (A) (A) and (B) only
- (B) (A), (B) and (D) only
- (C) (A) and (D) only
- (D) (A), (B), (C) and (D) only

Q59 Which of the following are examples of CSMA channel sensing methods?

- (A) 1-persistent
- (B) 2-persistent
- (C) p-persistent
- (D) o-persistent

Choose the correct answer from the options given below :

- (A) (A), (B) and (D) Only
- (B) (A), (C) and (D) Only
- (C) (B), (C) and (D) Only
- (D) (A), (B) and (C) Only

Q60 The selection of Spiral Model based on characteristics of requirements :

- (A) Are requirements easily understandable and

defined?

(B) Do we change requirements quite often?

(C) Can we define requirements early in the cycle?

(D) Requirements are indicating a complex to be built

Choose the correct answer from the options given below :

- (A) (C) Only
- (B) (B) Only
- (C) (B) and (D) Only
- (D) (A) and (C) Only

Q61 The work done by UDP is/are :

- (A) Congestion control
- (B) Flow control
- (C) Retransmission
- (D) Segment transmission

Choose the correct answer from the options given below :

- (A) (A) and (D) Only
- (B) (C) Only
- (C) (D) Only
- (D) (B) and (C) Only

Q62 Which of the following statement/s is/are NOT CORRECT?

- (A) OSPF is based on distance-vector routing protocol.
- (B) Both link-state and distance-vector routing are based on the least cost goal.
- (C) BGP4 is based on the path-vector algorithm.
- (D) The three-node instability can be avoided using split horizon combined with poison reverse.
- (E) RIP is based on link state algorithm.

Choose the correct answer from the options given below :

- (A) (A), (D) and (E) Only
- (B) (A) and (B) Only
- (C) (B) and (C) Only
- (D) (B), (C) and (E) Only

Q63 Which of the following statement/s are CORRECT?

- (A) NRZ is a bipolar scheme in which the positive voltage defined bit is 0 (zero).



(B) NRZ-L and NRZ-I both have an average signal rate of $N/2$.

(C) The idea of RZ and NRZ-L are combined into the Manchester scheme.

(D) NRZ-L and NRZ-I both have DC component problems.

(E) The minimum bandwidth of Manchester and differential Manchester is 3 times that of NRZ.

Choose the correct answer from the options given below :

(A) (A), (B) and (C) Only

(B) (A), (C), (D), and (E) Only

(C) (B), (C) and (D) Only

(D) (A), (B), (C), and (E) Only

Q64 Which of the following are commonly used parsing techniques in NLP (Natural Language Processing) for syntactic analysis?

(A) Top-down parsing

(B) Bottom Up parsing

(C) Dependency parsing

(D) Statistical machine translation

(E) Earley parsing

Choose the correct answer from the options given below :

(A) (A), (C), (D), (E) Only

(B) (B), (C), (D), (E) Only

(C) (A), (B), (C), (E) Only

(D) (A) and (B) Only

Q65 In the content of Alpha Beta pruning in game trees which of the following statements are correct regarding cut off procedures?

(A) Alpha Beta pruning can eliminate subtrees with certainty when the value of a node exceeds both the alpha and beta bonds.

(B) The primarily purpose of Alpha-Beta pruning is to save computation time by searching fewer nodes in the same tree.

(C) Alpha Beta pruning guarantees the optimal solution in all cases by exploring the entire game tree.

(D) Alpha and Beta bonds are initialized to negative and positive infinity respectively at the root node.

Choose the correct answer from the options given below :

(A) (A), (C), (D) Only

(B) (B),(C), (D) Only

(C) (A), (B), (D) Only

(D) (C), (B) Only

Q66 Match the List-I with List-II

(A) $\phi \cap \{\phi\} =$ (I) ϕ

(B) $\{\phi\} \cap \{\phi\} =$ (II) $\{\phi\}$

(C) $\{\phi, \{\phi\}\} - \phi =$ (III) $\{\{\phi\}\}$

(D) $\phi \cup \{\{\phi\}\} =$ (IV) $\{\phi, \{\phi\}\}$

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(B) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(D) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

Q67 Match List - I with List - II.

List - I

List - II

(a) SZA

(I) Increment M and skip

it zero

(b) SKI

(II) Skip if AC is negative

(c) SNA

(III) Skip if input flag is on

(d) ISZ

(IV) Skip if AC is Zero

Choose the correct answer from the options given below :

(A) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)

(B) (a)-(IV), (b)-(III), (c)-(II), (d)-(I)

(C) (a)-(IV), (b)-(II), (c)-(I), (d)-(III)

(D) (a)-(III), (b)-(IV), (c)-(II), (d)-(I)

Q68 Match List - I with List - II.

List - I	List - II
a. Bresenham	(I) Hidden surface removal
b. Cohen-Sutherland	(II) Line drawing algorithm
c. Sutherland-Hodgeman	(III) Line clipping algorithm
d. Z-Buffer	(IV) Polygon clipping algorithm

Choose the correct answer from the options given below :



- (A) (a)-(III), (b)-(II), (c)-(IV), (d)-(I)
 (B) (a)-(II), (b)-(III), (c)-(I), (d)-(IV)
 (C) (a)-(II), (b)-(III), (c)-(IV), (d)-(I)
 (D) (a)-(II), (b)-(IV), (c)-(III), (d)-(I)

Q69 Match List - I with List - II.

List - I	List - II
a. BCNF iff	(I) every JD is implied by the candidate keys
b. 5NF iff	(II) all underlying domains contain scalar values only
c. 1NF iff	(III) every MVD is implied by the candidate keys
d. 4NF iff	(IV) every FD is implied by the candidate keys

Choose the correct answer from the options given below :

- (A) (a)-(III), (b)-(II), (c)-(I), (d)-(IV)
 (B) (a)-(IV), (b)-(I), (c)-(II), (d)-(III)
 (C) (a)-(II), (b)-(III), (c)-(IV), (d)-(I)
 (D) (a)-(IV), (b)-(I), (c)-(III), (d)-(II)

Q70 Match List - I with List - II.

List-I	List-II
a. LRU	(I) A page is written to secondary memory only when it has been selected for replacement
b. Demand cleaning	(II) A page that has not been referenced for the longest time is replaced
c. Long-term scheduling	(III) The decision to add to the number of processes that are partially or fully in main memory
d. Medium term scheduling	(IV) The decision to add to the pool of processes to be executed

Choose the correct answer from the options given below :

- (A) (a)-(I), (b)-(III), (c)-(IV), (d)-(II)
 (B) (a)-(II), (b)-(I), (c)-(IV), (d)-(III)
 (C) (a)-(III), (b)-(II), (c)-(I), (d)-(IV)
 (D) (a)-(IV), (b)-(II), (c)-(III), (d)-(I)

Q71 Match List - I with List - II according to input to the compiler phase that process it :

List-I	List-II
a. Syntax tree	(I) Code generator
b. Intermediate representation	(II) Semantic analyzer
c. Token stream	(III) Lexical analyzer
d. Character stream	(IV) Syntax analyzer

Choose the correct answer from the options given below :

- (A) (a)-(IV), (b)-(III), (c)-(I), (d)-(II)
 (B) (a)-(II), (b)-(I), (c)-(IV), (d)-(III)
 (C) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)
 (D) (a)-(IV), (b)-(I), (c)-(II), (d)-(III)

Q72 Match List - I with List - II.

List-I	List-II
a. $P \wedge (P \rightarrow Q)$	I $P \vee Q$
b. $\neg(P \vee Q) \rightarrow (P \wedge Q)$	II $(P \wedge \neg P) \vee (P \wedge Q)$
c. $P \rightarrow Q$	III $(\neg P) \vee Q$
d. $P \vee (Q \wedge R)$	IV $(P \wedge P) \vee (P \wedge Q) \vee (P \wedge R) \vee (Q \wedge R)$

Choose the correct answer from the options given below :

- (A) (a)-(I), (b)-(II), (c)-(III), (d)-(IV)
 (B) (a)-(II), (b)-(I), (c)-(III), (d)-(IV)
 (C) (a)-(III), (b)-(I), (c)-(II), (d)-(IV)
 (D) (a)-(IV), (b)-(III), (c)-(II), (d)-(I)

Q73 Match List - I with List - II.

List-I	List-II
a. Greedy Best first search	I. The space complexity as $O(d)$ where d = depth of the deepest optimal solution
b. A^*	II. Incomplete even if the search space is



	finite
c. Recursive best-first search	III. Optimal if optimal solution is reachable otherwise return the best reachable optimal solution
d. SMA*	IV. Computation and space complexity is two light

Choose the correct answer from the options given below :

- (A) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)
 (B) (a)-(II), (b)-(III), (c)-(I), (d)-(IV)
 (C) (a)-(III), (b)-(II), (c)-(IV), (d)-(I)
 (D) (a)-(III), (b)-(IV), (c)-(II), (d)-(I)

Q74 Match List - I with List - II.

List-I	List-II
a. Monoalphabetic cipher	(I) Round key
b. DES	(II) One-to-many relationship
c. Stream Cipher	(III) One-to-one relationship
d. Polyalphabetic cipher	(IV) Feedback mechanism

Choose the correct answer from the options given below :

- (A) (a)-(III), (b)-(I), (c)-(IV), (d)-(II)
 (B) (a)-(III), (b)-(IV), (c)-(II), (d)-(I)
 (C) (a)-(I), (b)-(III), (c)-(IV), (d)-(II)
 (D) (a)-(II), (b)-(I), (c)-(IV), (d)-(III)

Q75 Match List - I with List - II.

List - I	List - II
a. Hill climbing	I. $O(b^d)$
b. Best first search	II. $O(bd)$
c. A* Search	III. $O(1)$
d. Depth first search	IV. $O(b^m)$

Choose the correct answer from the options given below :

- (A) (a)-(III), (b)-(I), (c)-(IV), (d)-(II)
 (B) (a)-(II), (b)-(I), (c)-(IV), (d)-(III)
 (C) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)

- (D) (a)-(I), (b)-(III), (c)-(II), (d)-(I)

Q76 Arrange the following in ascending order :

- (a) Remainder of 49^{16} when divided by 17
 (b) Remainder of 2^{446} when divided by 9
 (c) Remainder of 155^{17} when divided by 17
 (d) Last digits of the number 7^{45}

Choose the correct answer from the options given below :

- (A) (a), (b), (c), (d)
 (B) (a), (b), (d), (c)
 (C) (a), (c), (b), (d)
 (D) (d), (c), (b), (a)

Q77 In most general case, the computer needs to process each instruction with the following sequence of steps:

- (A) Calculate the effective address
 (B) Execute the instruction
 (C) Fetch the instruction from memory
 (D) Fetch the operand from memory
 (E) Decode the instruction

Choose the correct answer from the options given below :

- (A) (C),(E),(D),(A),(B)
 (B) (C),(E),(A),(D),(B)
 (C) (A),(B),(C),(E),(D)
 (D) (A),(B),(C),(D),(E)

Q78 Consider the following code segment :

```
int arr[]={0,1,2,3,4};
int i=1, *ptr;
ptr=arr+2;
```

arrange the following printf statements in the increasing order of their output.

- (a) `printf ("% d", ptr [i]);`
 (b) `printf ("%d", ptr [i+1]);`
 (c) `printf ("%d", ptr[-i]);`
 (d) `printf ("%d", ptr[-i+1]);`

Choose the correct answer from the options given below :

- (A) (c), (a), (b), (d)
 (B) (c), (d), (a), (b)
 (C) (d), (a), (b), (c)
 (D) (a), (b), (d), (c)



Q79 The steps for analysis and design of object oriented system.
 (a) Draw interaction diagrams
 (b) Draw state chart and object diagram
 (c) Draw use case and activity diagram
 (d) Draw component and deployment diagram
 (e) Draw class diagram
 Choose the correct answer from the options given below :

- (A) $(e) \rightarrow (b) \rightarrow (a) \rightarrow (c) \rightarrow (d)$
 (B) $(b) \rightarrow (a) \rightarrow (e) \rightarrow (d) \rightarrow (c)$
 (C) $(e) \rightarrow (c) \rightarrow (b) \rightarrow (d) \rightarrow (a)$
 (D) $(c) \rightarrow (a) \rightarrow (e) \rightarrow (b) \rightarrow (d)$

Q80 Arrange the following levels of interrupt protection within the Linux Kernel, in the order of increasing priority.
 (a) user mode programs
 (b) bottom half interrupt handlers
 (c) kernel system service routines
 (d) top half interrupt handlers
 Choose the correct answer from the options given below :

(A) (a),(b),(d),(c)
 (B) (a),(c),(b),(d)
 (C) (a),(c),(d),(b)
 (D) (d), (a), (c), (b)

Q81 Arrange the following phases of a compiler as per their order of execution (start to end)
 (a) Target code generation
 (b) Syntax Analysis
 (c) Code optimization
 (d) Semantic Analysis
 (e) Lexical Analysis
 Choose the correct answer from the options given below :

(A) (b), (e), (d), (a), (c)
 (B) (e), (d), (b), (a), (c)
 (C) (e), (b), (d), (c), (a)
 (D) (b),(d),(e),(a),(c)

Q82 The prototyping model has the sequence :
 (a) Customer Evaluation
 (b) Quick design

(c) Requirements
 (d) Implement
 (e) Design
 Choose the correct answer from the options given below :

- (A) $(c) \rightarrow (a) \rightarrow (d) \rightarrow (b) \rightarrow (e)$
 (B) $(b) \rightarrow (c) \rightarrow (a) \rightarrow (d) \rightarrow (e)$
 (C) $(c) \rightarrow (b) \rightarrow (d) \rightarrow (a) \rightarrow (e)$
 (D) $(e) \rightarrow (b) \rightarrow (c) \rightarrow (d) \rightarrow (a)$

Q83 Arrange the following encoding strategies used in Genetic Algorithms (GAs) in the correct sequence starting from the initial step and ending with the final representation of solutions :
 (a) Binary Encoding
 (b) Real valued Encoding
 (c) Permutation Encoding
 (d) Gray coding
 Choose the correct answer from the options given below :

(A) (d),(b),(a),(c)
 (B) (b),(d),(a),(c)
 (C) (c),(d),(a),(b)
 (D) (b),(c),(a),(d)

Q84 Arrange the following steps in the correct order for a DHCP Client to renew its IP lease with a DHCP server :
 (a) DHCP client sends a DHCPREQUEST message
 (b) DHCP server acknowledges the renewal with a DHCPACK message
 (c) DHCP client checks the local lease timer and initiates renewal
 (d) DHCP server updates its lease database
 Choose the correct answer from the options given below :

(A) (a),(b),(c),(d)
 (B) (c),(d),(b),(a)
 (C) (c),(b),(a),(d)
 (D) (c), (a),(b),(d)

Q85 Arrange the following steps in the correct sequence for applying an unsupervised learning technique such as K-means clustering to a data set:



- (a) Randomly initialize cluster centroids
- (b) Assign each data point to nearest cluster centroid
- (c) Update the cluster centroids based on the mean of data points assigned to each cluster
- (d) Specify the number of clusters (K) to partition the data into
- (e) Repeat steps B and C until convergence criteria are met

Choose the correct answer from the options given below :

- (A) (d),(a),(b),(c),(e)
- (B) (a),(b),(c),(d),(e)
- (C) (c),(b),(a),(d),(e)
- (D) (d),(c),(a),(b),(e)

Q86 Given below are two statements :

Statement (I): If H is a nonempty finite subset of a group G and $ab \in H \forall a, b \in H$ then H is also a group

Statement (II): There is no homomorphism existing from $(Z, +)$ to $(Q, +)$; where Z is a set of integers and Q is a set of rational numbers.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Q87 Given below are two statements :

Statement (I): The friend function and the member functions of a friend class directly access the private and protected data.

Statement (II): The friend function can access the private data through the member functions of the base class.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both Statement I and Statement II are correct

- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Q88 Given below are two statements :

Statement (I): A thread is a dispatchable unit of work that does not execute sequentially and is not interruptible

Statement (II): It is not possible to alter the behaviour of a thread by altering its context when thread is suspended

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Q89 Given below are two statements :

Statement (I) : In Reuse Oriented Model, Modification of the old system parts appropriate to the new requirements.

Statement (II) : In Reuse Oriented Model, Integration of the modified parts are not possible into the new systems.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Q90 Given below are two statements:

Statement (I) : In datagram networks - routers hold state information about connections



Statement (II) : In a virtual circuit network- each virtual circuit requires router table space per connection

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Both Statement I and Statement II are correct
 (B) Both Statement I and Statement II are incorrect
 (C) Statement I is correct but Statement II is incorrect
 (D) Statement I is incorrect but Statement II is correct

Directions (91-95) Read the following passage and answer the given questions.

Food X contains 6 units of Vitamin D per gram and 7 units of Vitamin E per gram and cost is Rs 12 per gram. Food Y contains 8 units of vitamin D per gram and 12 units of Vitamin E per gram and cost is Rs 20 per gram. The daily minimum requirements of vitamin D and E are 100 units and 120 units respectively.

Suppose x is the quantity (in gram) of food X, y is quantity (in gram) of food Y.

Answering the following question based on the above paragraph given.

Q91 The cost function of total food is:

- (A) $Z = 6x + 7y$ (B) $Z = 8x + 12y$
 (C) $Z = 12x + 20y$ (D) $Z = 20x + 12y$

Q92 Which of the following constraints when formulating the LPP?

- (A) $6x + 7y \leq 100, 8x + 12y \leq 120, x, y \geq 0$
 (B) $6x + 8y \leq 100, 7x + 12y \leq 120, x, y \geq 0$
 (C) $6x + 7y \geq 100, 8x + 12y \geq 120, x, y \geq 0$
 (D) $6x + 8y \geq 100, 7x + 12y \geq 120, x, y \geq 0$

Q93 Which of the following are quantities (in grams) of food X and Y respectively when the cost of food is minimum:

- (A) 0 and $12\frac{1}{2}$
 (B) 15 and $\frac{5}{4}$
 (C) $\frac{120}{7}$ and 0
 (D) 0 and 10

Q94 The dual of the formulated LPP is :

(A) Max $Z = 100u + 120v$

s.t.

$$6u + 7v \leq 12$$

$$8u + 12v \leq 20$$

$$u, v \geq 0$$

(B) Max $Z = 12u + 20v$

s.t.

$$6u + 7v \leq 100$$

$$8u + 12v \leq 120$$

$$u, v \geq 0$$

(C) Max $Z = 100u + 120v$

$$6u + 7v \leq 12$$

$$8u + 7v \leq 20$$

u, v are unrestricted

(D) Max $Z = 100u + 120v$

s.t.

$$6u + 7v \geq 12$$

$$8u + 12v \geq 20$$

$$u, v \geq 0$$

Q95 The minimum cost of food is:

- (A) 205 (B) 250
 (C) 330 (D) 200

Directions (96-100) Read the following passage and answer the given questions.

Consider the following relations $X(S, S_i, C)$ and $Y(S, P, D)$.

X

S	Si	C
J	1	M
B	2	N
R	3	H
T	4	G

Y

S	P	D
J	S1	CA
B	P1	AB
R	D1	DC
A	H1	MD

Q96



Find the number of tuples by applying the operation

$$X \bowtie X.S = Y.S^Y$$

- (A) 1 (B) 3
(C) 4 (D) 6

Q97 The result of the following operation will be

$$X \bowtie X.S = Y.S^Y$$

- (A)

S	Si	C	P	D
J	1	M	S1	CA
B	2	N	P1	AB
R	3	H	D1	DC
T	4	G	NULL	NULL
- (B)

S	Si	C	P	D
J	1	M	S1	CA
B	2	N	P1	AB
R	3	H	D1	DC
A	NULL	NULL	H1	MD
- (C)

S	Si	C	P	D
J	1	M	S1	CA

B	2	N	P1	AB
R	3	H	D1	DC
T	4	G	NULL	MD

(D) None of these

- Q98** Which of the following join is used to get all the tuples of relation X and Y with Null values of corresponding missing values?
(A) Left outer join
(B) Right outer join
(C) Natural join
(D) Full outer join

- Q99** Number of tuples by applying right outer join on relation X and Y is/are :
(A) 16 (B) 5
(C) 3 (D) 4

- Q100** Number of tuples obtained by applying cartesian product over X and Y are :
(A) 16 (B) 12
(C) 04 (D) 32



Answer Key

Q1 (B)
Q2 (D)
Q3 (A)
Q4 (B)
Q5 (C)
Q6 (C)
Q7 (C)
Q8 (D)
Q9 (B)
Q10 (C)
Q11 (C)
Q12 (D)
Q13 (C)
Q14 (D)
Q15 (D)
Q16 (B)
Q17 (B)
Q18 (A)
Q19 (C)
Q20 (B)
Q21 (C)
Q22 (C)
Q23 (C)
Q24 (D)
Q25 (B)
Q26 (C)
Q27 (D)
Q28 (D)
Q29 (A)
Q30 (C)
Q31 (C)

Q32 (C)
Q33 (C)
Q34 (C)
Q35 (D)
Q36 (C)
Q37 (C)
Q38 (C)
Q39 (D)
Q40 (A)
Q41 (D)
Q42 (B)
Q43 (D)
Q44 (C)
Q45 (B)
Q46 (C)
Q47 (B)
Q48 (B)
Q49 (B)
Q50 (B)
Q51 (D)
Q52 (D)
Q53 (C)
Q54 (B)
Q55 (B)
Q56 (C)
Q57 (D)
Q58 (A)
Q59 (B)
Q60 (C)
Q61 (C)
Q62 (A)



Q63 (C)
Q64 (C)
Q65 (C)
Q66 (D)
Q67 (B)
Q68 (C)
Q69 (B)
Q70 (B)
Q71 (B)
Q72 (B)
Q73 (A)
Q74 (A)
Q75 (A)
Q76 (C)
Q77 (B)
Q78 (B)
Q79 (D)
Q80 (B)
Q81 (C)

Q82 (C)
Q83 (C)
Q84 (D)
Q85 (A)
Q86 (C)
Q87 (A)
Q88 (B)
Q89 (C)
Q90 (D)
Q91 (C)
Q92 (D)
Q93 (B)
Q94 (A)
Q95 (A)
Q96 (C)
Q97 (B)
Q98 (D)
Q99 (D)
Q100 (A)



Hints & Solutions

Q1 Text Solution:

Let E_1 be the event that the integer is divisible by 2 and E_2 be the event that it is divisible 5. Then the event that the integer is divisible by 2 or 5 is $E_1 \cup E_2$ and $E_1 \cap E_2$ is the event that it is divisible by 2 and 5. It follows that:

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2) \text{ ----- (1)}$$

Here, a positive integer not exceeding 100, so the total number of outcomes will be 100

From 1 to 100, there are a total of 50 numbers that are divisible by 2, so favorable number of outcomes $E_1 = 50$

From 1 to 100, there are a total of 20 numbers that are divisible by 5, so favorable number of outcomes $E_2 = 20$

From 1 to 100, there are a total of 10 numbers that are divisible by 2 and 5, so favorable number of outcomes $E_1 \cap E_2 = 10$

So from equation (1)

$$p(E_1 \cup E_2) = 50/100 + 20/100 - 10/100 = 3/5.$$

Therefore option B is the correct answer.

Q2 Text Solution:

A string is called a palindrome if the reverse of the string is the same as the original one.

Option A, B, and C are palindrome as their sequences remain unchanged when reversed and these are also subsequences of the given string ababcdabba.

Here string $\{a, 'd', b, 'a'\}$ is not a palindromic subsequence of the string "ababcdabba", because the reverse of the string adba is not the same as the string itself.

So option D is the correct answer.

Q3 Text Solution:

AVL trees and binary search trees are two commonly used data structures for storing and searching data. Both these data structures are binary trees, which means that each node can have at most two child nodes. However, AVL trees are self-balancing binary search trees, while binary search trees may or may not be balanced.

Searching in AVL Trees and Binary Search Trees:

Searching for a value in a binary search tree or an AVL tree involves traversing the tree from the root node to a leaf node that contains the value.

The cost of searching a tree is determined by the number of nodes that must be visited to find the value. In a balanced tree, the cost of searching is proportional to the height of the tree.

The Cost of Searching AVL Trees and Binary Search Trees: The cost of searching an AVL tree is logarithmic, which means that the number of nodes that must be visited to find a value is proportional to the logarithm of the number of nodes in the tree. The cost of searching a binary search tree is also logarithmic if the tree is balanced. However, if the binary search tree is unbalanced, the cost of searching may be linear, which means that the number of nodes that must be visited may be proportional to the number of nodes in the tree.

The cost of searching an AVL tree is $\theta(\log n)$ but that of a binary search tree is $O(n)$ if the binary search tree is unbalanced. However, if the binary search tree is balanced, the cost of searching is also $\theta(\log n)$. AVL trees are always balanced, so the cost of searching is guaranteed to be logarithmic. Therefore, the correct answer is option 'A'.

Q4 Text Solution:

Postfix expression " $43*25*+8-$ "

Push: 4 and 3 (Top of stack)

Multiply: $4*3 = 12$

Push: 12 2 and 5 (Top of stack)

Multiply: $2*5 = 10$

Push 10

Add: $12+10 = 22$

Push 22 and 8

Subtract: $22-8 = 14$

Therefore option B is the correct answer.

Q5 Text Solution:



- An encoder is a combinational circuit that converts binary information in the form of 2^N input lines into N output lines, which represent N bit code for the input. Decoders decode the N bits into 2^N lines.
- A Flip-Flop is a sequential circuit and stores 1-bit data. It has two stable states, One of its two states represents a "one" and the other represents a "zero".
- A register is a small and fast storage memory within the CPU or the ALU of a computer. Each register can store 8 bits. To store more than 8 bits, we have to use two registers in pairs.

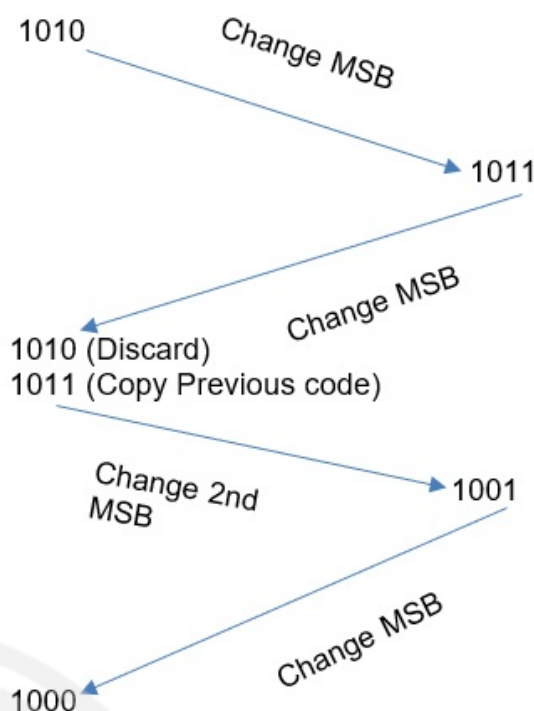
So option C is the correct answer.

Q6 Text Solution:

Gray code is a binary numbering system in which two successive values only differ by one bit is called gray code.

The prefix and reflect method are recursively used to generate the Gray code of a number. For generating gray code:

1. We find the number of bits required to represent a number.
2. Next, we find the code for 0, i.e., 0000, which is the same as binary.
3. Now, we take the previous code, i.e., 0000, and change the most significant bit of it.
4. We perform this process recursively until all the codes are uniquely identified.
5. If by changing the most significant bit, we find the same code obtained previously, then the second most significant bit will be changed, and so on.



So the correct answer is C.

BCD is a binary-encoded representation of decimal numbers. In this, each decimal digit is represented by a group of 4 binary digits or bits in much the same way as Hexadecimal. So for the 10 decimal digits (0-to-9), we need a 4-bit binary code.

The excess-3 code is a BCD code used to represent the decimal numbers.

We can find the excess-3 code of the given binary number by using the following steps:

We find the decimal number of the given binary number.

Then we add 3 in each digit of the decimal number.

Now, we find the binary code of each digit of the newly generated decimal number.

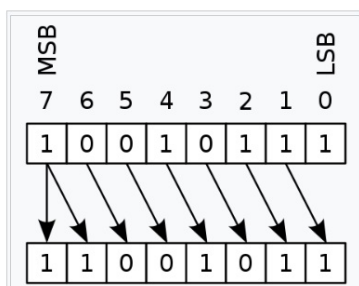
We can also add 0011 in each 4-bit BCD code of the decimal number for getting an excess-3 code.

There is no widely recognized coding system as Excess-3 Gray code.

Q7 Text Solution:

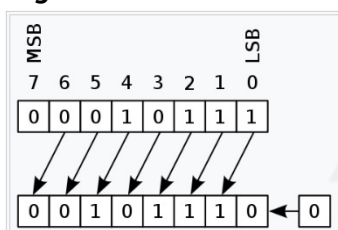
Arithmetic shift right - It moves bits in one position to the right.





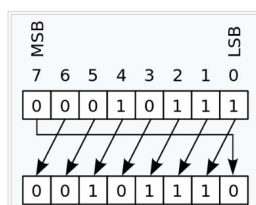
The empty position in the most significant bit is filled with a copy of the original MSB. So the leftmost bit (sign bit) stays the same to preserve the sign of the number. So a positive number remains a positive and a negative number remains negative. So an arithmetic right shift can divide a signed binary number by 2.

Logical Shift



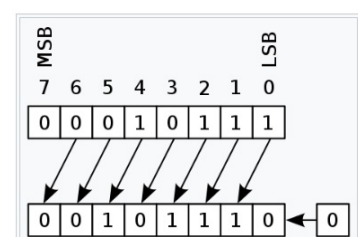
The number of bit positions in a given value shall be shifted, such as a shift left by 1, and the empty position in the least significant bit is filled with 0. A logical shift does not preserve a number's sign bit.

Circular shift



A rotation (or circular shift) is an operation in which the bits that fall off at one end are put back to the other end. This maintains the number of bits in data does not divide the number.

Arithmetic shift left



A left arithmetic shift of a binary number by 1. The empty position in the least significant bit is filled with a zero which multiplies the number by 2.

So option C is the correct answer.

Q8 Text Solution:

Option A, Micro-program: A sequence of micro-instructions is called a micro-program, which is stored in a ROM or RAM called a Control Memory (CM).

Option B, Micro-operations are low-level instructions that perform basic operations on data stored in one or more registers, including transferring data between registers or between registers and external buses of the central processing unit (CPU) and performing arithmetic or logical operations on registers.

Option C, A routine is nothing but a sequence of code, intended for the execution of user programs and input/output operations. It is called repeatedly by other codes, during the execution of a program.

Option D, Subroutines are programs that are used by other routines to accomplish a particular task. A subroutine can be called from any point within the main body of the micro-program.

Frequently, many micro-programs contain identical sections of code. Microinstructions can be saved by employing subroutines that use common sections of microcode.

For example, the sequence of micro-operations needed to generate the effective address of the operand for instruction is common to all memory reference instructions. This sequence could be a subroutine that is called from within many other routines to execute the effective address computation.

So Option D is the correct answer.

Q9 Text Solution:

To magnify a shape, we can use the principle of scalings in transformations. We keep the points R(10,4) fixed and magnify everything with a factor of 4 from that point.



Start with the point P(0,0)

1. The vector from R to P is $P-R=(0,0)-(10,4)=(-10,-4)$.

2. Now, we multiply this vector by 4 (our scale factor) to get $(-40,-16)$.

3. Then, we add this to our fixed point R, to get the new position for P, $R+4x(P-R)=(10,4)+(-40,-16)=(-30,-12)$.

Similarly, for point Q(2,2).

1. The vector from R to Q is $Q-R=(2,2)-(10,4)=(-8,-2)$.

2. We multiply this vector by scale factor 4, to get $(-32,-8)$.

3. Adding this to R, we find the new position for Q: $R+4x(Q-R)=(10,4)+(-32,-8)=(-22,-4)$.

The coordinates of the magnified triangle are thus P(-30,-12), Q(-22,-4), and R(10,4).

So option B is the correct answer.

Q10 Text Solution:

A function that is declared in a base class but not implemented there is known as a pure virtual function. Rather, it needs to be implemented in any derived class that isn't abstract.

A pure virtual function often has the following syntax:

virtual void print() = 0 ;

Option A) Virtual functions do not implement polymorphism:

This statement is false. Virtual functions are a fundamental feature of polymorphism in object-oriented programming. Polymorphism allows objects of different types to be treated as objects of a common base type, and virtual functions enable dynamic method binding, which is essential for achieving polymorphic behavior.

Option (B) Virtual functions do not permit calling of derived class functions using a base class pointer:

This statement is false. Virtual functions allow derived class functions to be invoked through a base class pointer or reference.

Option C) We can never build an object from a class containing a pure virtual function

Classes classified as abstract have one or more pure virtual functions. The aim of an abstract class is to provide a suitable base class from which other classes can inherit. Abstract classes can be seen as an interface and cannot be utilized to instantiate objects. A compilation error occurs when you try to instantiate an object of an abstract class.

Option D) Pure virtual function can never have a body

This statement is false. Pure virtual functions can have a body, but it's optional. When a pure virtual function has a body, it provides a default implementation that can be overridden by derived classes.

So option C is the correct answer.

Q11 Text Solution:

In the above program-

- The variable i is initialized to 3. After executing the while(i--), the value of i=2.
- inside the inner loop again the i is initialized to 10, which is decremented by 1 (10-1=9), so for the first print statement 9 will be printed.
- Next for i=2, while(i--)) will be executed and inner statements 9 will be printed.
- Next for i=1, while (i--)) will be executed and inner statements 9 will be printed.
- Next for i=0, the condition for the while loop will be false. and will come out from the while loop. Here the value of i = -1 will be printed.

After executing the program 999-1 will be printed.

So option C is the correct answer.

Q12 Text Solution:

Option A, Luminance:

1. This is a photometric measure of the luminous intensity per unit area of light traveling in a given direction.
2. It describes the amount of light that passes through or is emitted from a particular area and falls within a given solid angle. It's often used in



the context of the emission or reflection of light from flat, diffused surfaces.

Option B, Saturation:

1. In color theory, saturation refers to the intensity of a color. It's the degree to which a color is different from a gray of the same brightness or lightness.
2. A highly saturated color is vivid, while an unsaturated color appears washed out or muted.

Option C, Incandescence:

1. This is the emission of light from a hot body as a result of its temperature. The color of incandescent light varies with the temperature of the light-emitting body.
2. This term is often used to describe the light emitted by traditional light bulbs, which work by heating a filament until it glows. Incandescence mainly connects to temperature.

Option D, Wavelength

1. This is the distance between two successive crests or troughs in a light wave.
 2. The wavelength of light determines its color or hue. Different colors of light have different wavelengths.
 3. For instance, violet light has a short wavelength, while red light has a long wavelength. So this is the term most closely connected with the hue of a color.
- So option D is the correct answer.

Q13 Text Solution:

Option A, Normalization is the process of structuring the data in a database is called normalization. Redundancy from a relation or collection of relations is reduced through the use of normalization. Additionally, it is used to eliminate undesirable characteristics including Update, Insertion, and Deletion Anomalies. The bigger table is divided into smaller ones by normalization, and relationships are used to connect them.

Option B, The set of commands known as the Data Manipulation Language, or DML is used for modifying, adding, or removing rows from SQL tables to manipulate data in a database.

Option C, Data Definition Language (DDL) is used to create and alter the structure of objects within a database by using preset commands and a particular syntax. Tables, sequences, locations, aliases, schemas, and indexes are some examples of these database objects.

Option D, Primary key is an attribute or a set of attributes. One tuple can be distinguished from every other tuple in the relation using the primary key. It assists the database system in locating and identifying a specific tuple within the relation.

Q14 Text Solution:

- A memory address is a reference to a specific memory location used at various levels by software and hardware.
- An address that is created by the CPU while a program is running is known as a logical address. Because the logical address is virtual that is, it doesn't exist physically—it is also referred to as a virtual address. The CPU uses this address as a reference to go to the physical memory location.
- A physical address is the actual address in the main memory where data is stored. It is a location in physical memory, as opposed to a virtual address. Physical addresses are used by the Memory Management Unit (MMU) to translate logical addresses into physical addresses.
- The fundamental difference between logical and physical addresses is that the CPU generates the logical address during program execution, on the other hand, the physical address refers to a location in the memory unit.

So option D is the correct answer.

Q15 Text Solution:

Option A, MIMD (Multiple Instruction, Multiple Data):

Multiple processors can simultaneously execute different instructions on separate data in MIMD. It's a feature of more sophisticated



multiprocessing or parallel computing architectures.

Option B, DMA (Direct Memory Access):

DMA is a technology that allows I/O devices or hardware devices to transfer data between themselves and main system memory independently of the central processing unit (CPU). It's used to speed up data transfer and improve system performance.

Option C, Asynchronous Bus:

A bus that links the components of a computer system so that data flows are self-timed as opposed to regulated by a synchronizing clock signal.

Option D, Synchronous Bus:

This is a computer bus where a clock signal controls the regular intervals at which data is sent. Efficient data transmission is possible since the source and destination are aware of these intervals simultaneously.

So option D is the correct answer.

Q16 Text Solution:

A macro is a name given to a block of C statements as a pre-processor directive. In the given program a macro is defined so before our program compiles, the macro will be processed.

Here

$a = \text{SQR}(b+2)$; is given and value of $b=3$

So $\text{SQR}(3) = 3*3=9$ will be executed first then 2 will be added to 9 i.e. output will be 11.

So option B is the correct answer.

Q17 Text Solution:

SCAN Disk Scheduling Algorithm

Step 1 – The array has the elements that have requested resources from different processes in ascending order based on arrival time.

Step 2 – The disk arm starts from one end of the disk and moves towards the other end.

Step 3 – When the disk moves in this direction, it will service all tracks one by one based on the request.

Step 4 – Then calculate the total distance traveled from the head.

Step 5 – The new position is identified by the currently serviced request

Step 6 – Then step 3 is repeated when it reaches one end of the disk.

Step 7 – When the endpoint is reached, it will reverse the direction and go back to step 2 when the entire request has been serviced.

In the given question, the head has just finished a request at track 125 and is currently at track 143.

That means it is moving towards a higher number of tracks.

So we will take the difference between the highest track number and the current track number. i.e. $199-143=56$.

Then for the reverse direction we will take the difference between the highest track number and smallest track number i.e. $199-86=113$

Then we will add both the differences i.e. $56+113=169$ cylinders

Q18 Text Solution:

Number of virtual pages=virtual space/ individual page size

$= 4.3 \times 10^9 / 4 \times 10^3 \text{ bytes} = 1,075,000 \text{ pages}$

Size of the whole table = Number of virtual pages x each page table entry

$= 1,075,000 \times 5 = 5,375,000 \text{ bytes} = 5375 \text{ KB}$

Q19 Text Solution:

Option A) / etc/password

This option is not valid in Linux.

Option B) / root/password

This directory and file combination doesn't typically exist in a default Linux system

Option C) /etc/passwd is a plain text-based database that contains information for all user accounts on the system. It is owned by root.

In newer Linux, the hashed passwords are stored in the /etc/shadow file. In turn, the hashed password field in the /etc/passwd file is filled with the x character.

Option D) / root/passwd

This directory and file combination doesn't typically exist in a default Linux system.

So option C is the correct answer.

Q20 Text Solution:



Option A, Public Key: A public key and a private key are the two keys used in public-key cryptography. While the private key is kept secret and used for decryption, the public key is used for encryption and shared with others. Only the matching private key can decrypt messages that have been encrypted using the public key.

Option B, Digital signatures are a type of cryptography that guarantees the integrity and authenticity of a message or document. In order to confirm that the message has not been altered and was, it uses the sender's private key to encrypt a hash (message digest) of the content. The recipient can use the sender's public key to decrypt the signature which is signed by sender.

Option C, The output of the encryption process is known as the cipher text. When the original message is encrypted, a communication becomes unreadable cipher text. This procedure usually involves transforming the communication into a secure, coded format using a symmetric or asymmetric key algorithm.

Option D, The Data Encryption Standard (DES) is a symmetric-key data encryption algorithm. It only requires one key to be used for encryption and decryption. There are several iterations of permutation and substitution in this procedure. So option B is the correct answer.

Q21 Text Solution:

Option A, The combination of {ab, aa, baa} forms abaabaaabaa

ab aa baa ab aa

Option B, The combination of {ab, aa, baa} forms aaaabaaaa

aa aa baa aa

Option C, The combination of {ab, aa, baa} does not form baaaaabaaaab

baa aa ab aa aab (here aab is not in the combination)

Option D, The combination of {ab, aa, baa} forms baaaaabaa

baa aa ab aa

So option C is the correct answer.

Q22 Text Solution:

1. $A = \{a, b\}$

2. $L = A^*$ (the Kleene closure of A, which includes all possible strings over A including the empty string)

3. $X = \{a^n b^n \mid n > 0\}$ (the set of strings with an equal number of 'a's and 'b's, and at least one 'a' and one 'b')

Now take

$L \cup X = A^*$ (includes an equal number of 'a's and 'b's including the empty string)

It is a subset of A^* , so this is a regular language with a regular expression $(a,b)^*$

Now for $X = \{a^n b^n \mid n > 0\}$

This language consists of strings of an equal number of 'a's and 'b's, with at least one 'a' and one 'b'. Whenever unbounded storage is required for storing the count and then comparison with other unbounded counts, then the language is not regular. Since regular languages are languages that can be generated from one-element languages by applying certain standard operations a finite number of times.

So option C is the correct answer.

Q23 Text Solution:

For lookup, insertion, and deletion operations, hash tables have an average-case time complexity of $O(1)$. Yet, these operations may, in the worst case, require $O(n)$ time, where n is the number of elements in the table.

Q24 Text Solution:

The correct option is $D \mid n, E + n$ and $E \times n$

$E \rightarrow E + n \mid E \times n \mid n$

Input String $n + n \times n$

$\Rightarrow n + n \times n$

$\Rightarrow E + n \times n$ reduction $E \rightarrow n$

$\Rightarrow E \times n$ reduction $E \rightarrow E + n$

$\Rightarrow E$ reduction $E \rightarrow E \times n$

So the reductions are $n, E + n, E \times n$

Q25 Text Solution:

Option A, A Linked List is a linear data structure used for storing a collection of elements. Unlike arrays, linked lists use nodes to store elements that are not stored in contiguous memory



locations. Some of examples where the linked list is used, to implement stacks and queues, symbol table.

Option B, A hash table is typically an array of linked lists.

$\text{index} = f(\text{key}, \text{array_size})$

When we want to insert a key/value pair, we first need to use the hash function to map the key to an index in the hash table. Given a key, the hash function can suggest an index where the value can be found or stored:

$\text{hash} = \text{hashfunc}(\text{key})$

$\text{index} = \text{hash} \% \text{array_size}$

Option C, A binary tree is a tree that has at most two children for any of its nodes. Binary trees can be used to

1. store data in a database system, with each node representing a record.
 2. Also, it can be used to implement file systems, where each node represents a directory or file.
- This allows for efficient navigation and searching of the file system.

Option D, A stack is a linear data structure in which the insertion of a new element and removal of an existing element takes place at the same end represented as the top of the stack.

Some of the examples-

1. Stacks are used to keep track of the return addresses of function calls, allowing the program to return to the correct location after a function has finished executing.
2. Stacks are used to allocate and manage memory in some operating systems and programming languages.

So option B is the correct answer.

Q26 Text Solution:

Option A, With linear probing, every hash table cell is viewed as a slot with the capacity to hold a single key-value pair. This is an open-addressing technique.

The algorithm looks through the table linearly until it finds an empty slot when a collision happens. One drawback of linear probing is that

it may result in clustering, which is the filling of a group of nearby cells, hence lengthening the average search time.

Option B, Quadratic probing is an open addressing method designed to handle collisions, much like linear probing. But upon collision, rather than looking for an empty slot linearly. Quadratic probing uses a quadratic function $h_i(\text{value}) = (h(\text{value}) + i^2) \% \text{size}$, where i is the attempt number.

Option C, The process of implementing each cell in the Hash table as a distinct linked list or any other dynamic data structure is known as chaining.

In the event of a collision, the key-value pair is kept in the same slot as the linked list's subsequent node. This eliminates the need for a space in the hash table to manage collisions and offers a simple and effective method of doing so. The drawback, though, is the possibility of inefficiency brought on by a mishandled key distribution or collided keys. It can cause some chains to get extremely long while others get short or empty.

Option D, Double hashing, a collision resolution technique in hash tables, that uses two hash functions to compute two different hash values for a key. This method is less susceptible to clustering than linear probing and quadratic probing but more computation is required to calculate two hash functions.

So option C is the correct answer.

Q27 Text Solution:

	$n = 256$
$f(n) = 3n^{\sqrt{n}}$	3×256^{16} $= 3 \times 2^{128}$
$g(n) = 2^{\sqrt{n} \log_2 n}$	$2^{16 \times 8}$ $= 2^{128}$
$h(n) = n!$	$256!$ $= O((2^8)^{256})$ $= O(2^{2048})$

here $f(n)$ and $g(n)$ are having same order growth as $f(n) = 3 \times g(n)$. i.e. $f(n)$ is $O(g(n)) < h(n)$

So option D is the correct answer.



Q28 Text Solution:

Option A) An AVL tree is defined as a self-balancing Binary Search Tree (BST) where the difference between heights of left and right subtrees for any node cannot be more than one.

Option B) AA trees are the variation of the red-black trees. AA trees use the concept of levels to balance binary trees. The level of node (instead of color) is used for balancing information.

Option C) A 2-3 tree is a B-tree of order 3. Nodes with two children are called 2-nodes. The 2-nodes have one data value and two children. Nodes with three children are called 3-nodes. The 3 nodes have two data values and three children.

Option D) Red-black trees and 2-3-4 trees are comparable data structures since they are an isometry. There is at least one red-black tree with data items in the same order for every 2-3-4 trees.

Furthermore, color-flipping and rotations in red-black trees are analogous to insertion and deletion operations on 2-3-4 trees that result in node expansions, splits, and merges. Red-black tree introductions typically start with 2-3-4 trees since they are conceptually easier to understand. So option D is the correct answer.

Q29 Text Solution:

Option A, The average time required to search a storage location in memory and obtain its contents is called access time.

Option B, Latency time is the time required by the read/write head to move from one sector to another.

Option C, Response time is the time spent when the process is in the ready state and gets the CPU for the first time.

Option D, Reading time is the time spent in a system call that specifies the name of the file and finds where in the memory the next block of the file is for reading.

So option A is the correct answer.

Q30 Text Solution:

Option A) Make parsing and semantic analysis simpler- In the second phase of the compiler design parsing is done and in the third phase semantic analysis is done.

Option B) Improve error recovery and error reporting- At semantic analysis, the compiler checks the meaning of the source code to ensure that it makes sense. The compiler performs type checking, which ensures that variables are used correctly and that operations are performed on compatible data types. The compiler also checks for other semantic errors, such as undeclared variables and incorrect function calls.

Option C) When we generate intermediate code it is independent of the machine so we can easily implement the same code in different machines without performing the analysis phase of the compiler again and again.

Option D) Register allocation is a crucial optimization step in code generation, where the compiler assigns variables and values to processor registers to minimize memory access and improve execution speed.

So option C is the correct answer.

Q31 Text Solution:

- DFA refers to deterministic finite automata. Deterministic refers to the uniqueness of the computation. The finite automata are called deterministic finite automata if the machine reads an input string one symbol at a time. In DFA, there is only one path for specific input from the current state to the next state.
- DFA does not accept the null move, i.e., the DFA cannot change state without any input character. DFA can contain multiple final states.

The minimum number of final states for a DFA is 1. Therefore, $\text{Min}=1$.

The maximum number of final states for a DFA can be n . In this case, the final state is P .

Therefore, $\text{Max} = P$.

Sum of Minimum and Maximum Number of Final States: $\text{Min} + \text{Max} = P + 1$, So option C is the correct



answer.

Q32 Text Solution:

Test suite is a container that has a set of tests that helps testers in executing and reporting the test execution status. It can take any of the three states namely Active, Inprogress and completed.

Q33 Text Solution:

As each frame contains 2 bits per channel and 4 channels are there of 100Kbps

So link carries $100 \text{ kbps} / 2 = 50000$ frames per sec

Bit rate = frame rate x no. of bits per frame
 $= 50000 \times 8 = 400 \text{ kbps}$

So option C is the correct answer.

Q34 Text Solution:

TCP Header				
Bits	0-15			16-31
0	Source port			Destination port
32	Sequence number			
64	Acknowledgment number			
96	Offset	Reserved	Flags	Window size
128	Checksum			Urgent pointer
160	Options			

Option A, Sequence number: The order of the bytes sent in a TCP connection is indicated by this field in the TCP header. This information is essential for handling retransmissions and making sure the data is received in the right order.

Option B, Checksum: The TCP header's checksum field allows for error-checking, assisting in verifying the accuracy of the data and TCP header. Based on the data and TCP header contents, the sender computes the checksum, which the recipient confirms to identify any transmission errors.

Option C, Fragmentation offset: As shown in the figure, fragmentation offset is not a field in TCP header, it belongs to IP header. Large packets can be broken up into smaller fragments by the IP protocol in order to be sent over networks with smaller Maximum Transmission Unit (MTU) sizes. The location of a fragment in the original, unfragmented packet is indicated by the fragmentation offset field in the IP header.

Option D, Window size: Flow control in a TCP connection is managed by the window size field

in the TCP header. It shows the maximum amount of data (in bytes) that can be delivered by the sender before the recipient must acknowledge it. Congestion control and effective data transport are made possible by adjusting the window size. So option C is the correct answer.

Q35 Text Solution:

frames in bit-stuffing protocols start and end with the **0x7E** pattern, and extra 0s are inserted during transmission to avoid confusion with this delimiter

So option D is the correct answer

Q36 Text Solution:

Option A, A system's dynamic behavior is shown in a use case diagram. It incorporates use cases, actors, and their interactions to encapsulate the functionality of the system. It simulates the operations, services, and tasks needed by a system or application subsystem. It illustrates a system's high-level functionality and describes how a user interacts with it.

Option B, UML Sequence Diagrams are interaction diagrams that show the steps involved in operating. They depict how items interact with one another when working together. Sequence diagrams are time-focused and visually depict the sequence of the exchange by displaying the time, what messages are transmitted, and when on the vertical axis of the diagram.

Option C, At the highest level, Level-0 Data Flow Diagram(DFD) offers a comprehensive picture of the entire system. It does not go into detail on how these processes are internally operated; instead, it only displays the main data flows, stores, and processes in the system.

Another name for it is a context diagram. The perspective is intended to be abstract, presenting the system as a unified process with a connection to outside entities.

Option D, The Prototype Design Pattern enables the creation of new objects by copying an existing object. Prototype allows us to hide the complexity of making new instances from the



client. The concept is to copy an existing object rather than create a new instance from scratch, something that may include costly operations. The existing object acts as a prototype and contains the state of the object.
So option C is the correct answer.

Q37 Text Solution:

Option A) In the context of artificial intelligence and optimization, a "single agent with multiple objectives" refers to a scenario where an agent (such as an AI system or algorithm) is tasked with optimizing multiple conflicting objectives simultaneously.

Option B) In contrast to a single agent with multiple objectives, a "multi-agent system with a single objective" involves multiple agents working collaboratively towards achieving a common goal or objective.

Option C) Multiple intelligent agents that interact with their surroundings and one another to accomplish individual or group objectives make up Multi-Agent Systems (MAS). To differentiate itself from single-agent and distributed systems, MAS is characterized by autonomy, interaction, goal orientation, and distributed control.

Option D) A multiagent system with two objectives refers to a scenario where multiple agents are involved, and the overall system aims to optimize or achieve two distinct objectives simultaneously.

So option C is the correct answer.

Q38 Text Solution:

Fuzzy sets are sets whose elements have degrees of membership.

The membership function $\mu_A(x)$ of a fuzzy set A is a function $\mu_A : X \rightarrow [0,1]$

So, every element in x in X has membership degree: $\mu_A(x) \in [0,1]$

A is completely determined by the set of tuples:

$$A = \{(x, \mu_A(x)) \mid x \in X\}$$

Option A, The core of a fuzzy set A is the set of all points $x \in X$ such that $\mu_A(x) = 1$

$$\text{Core}(A) = \{x \mid \mu_A(x) = 1, x \in X\}$$

Option B, The support of a fuzzy set A is the set of all points $x \in X$ such that $\mu_A(x) > 0$

$$\text{Support}(A) = \{x \mid \mu_A(x) > 0, x \in X\}$$

Option C, A crossover point of a fuzzy set A is a point $x \in X$ at which $\mu_A(x) = 0.5$

$$\text{Crossover}(A) = \{x \mid \mu_A(x) = 0.5\}$$

Option D, The α -cut of a fuzzy set A is a crisp set defined by $A_\alpha = \{x \mid \mu_A(x) \geq \alpha\}$

So option C is the correct answer.

Q39 Text Solution:

To find the fitness value of each individual in the population, we can use the given fitness function

$$f(x) = x^2 - 4x + 4$$

Let's calculate the fitness values for the individuals with the given values of x:

1. For $x=1.5$

$$f(1.5) = (1.5)^2 - 4(1.5) + 4 = 2.25 - 6 + 4 = 0.25$$

2. For $x=2.0$

$$f(2.0) = (2.0)^2 - 4(2.0) + 4 = 4 - 8 + 4 = 0$$

3. For $x=3.0$

$$f(3.0) = (3.0)^2 - 4(3.0) + 4 = 9 - 12 + 4 = 1$$

4. For $x=4.5$

$$f(4.5) = (4.5)^2 - 4(4.5) + 4 = 20.25 - 18 + 4 = 6.25$$

So, the fitness values for the individuals are:

$$\{0.25, 0, 1, 6.25\}$$

The individual with the highest fitness value is $x=4.5$ with a fitness value of 6.25.

So option D is the correct answer.

Q40 Text Solution:**Input to Hidden Layer**

Each hidden neuron takes all 4 input values, multiplies them with its weights, then adds them.

Hidden neuron 1 (H1):

$$H1 = 0.1(0.5) + 0.3(0.8) + 0.5(0.2) + 0.2(0.6) = 0.51$$

Apply sigmoid:

$$A1 = \frac{1}{1 + e^{-0.51}} \approx 0.624$$

Hidden neuron 2 (H2):



$$H2 = 0.2(0.5) + 0.4(0.8) + 0.6(0.2) + 0.2(0.6) = 0.66$$

Sigmoid:

$$A2 = \frac{1}{1 + e^{-0.66}} \approx 0.658$$

Hidden neuron 3 (H3):

$$H3 = 0.3(0.5) + 0.5(0.8) + 0.7(0.2) + 0.2(0.6) = 0.81$$

Sigmoid:

$$A3 = \frac{1}{1 + e^{-0.81}} \approx 0.692$$

So hidden layer outputs are:

$$[A1, A2, A3] = [0.624, 0.658, 0.692]$$

Hidden to Output Layer

Now we use these hidden outputs as inputs to the output layer.

Output neuron 1 (O1):

$$O1 = 0.4(A1) + 0.1(A2) + 0.3(A3)$$

$$O1 = 0.4(0.624) + 0.1(0.658) + 0.3(0.692) = 0.523$$

Sigmoid:

$$Y1 = \frac{1}{1 + e^{-0.523}} \approx 0.628 \approx 0.62$$

Output neuron 2 (O2):

$$O2 = 0.5(A1) + 0.2(A2) + 0.4(A3)$$

$$O2 = 0.5(0.624) + 0.2(0.658) + 0.4(0.692) = 0.72$$

Sigmoid:

$$Y2 = \frac{1}{1 + e^{-0.72}} \approx 0.675 \approx 0.68$$

The **output layer values** are:

$$[0.62, 0.68]$$

Hence, the Correct answer is Option A.

Q41 Text Solution:

(A) $\exists x \forall y (x+y=y)$: This means there exists some real number x such that for all real numbers y , $x+y=y$. This statement is true for $x=0$. so (A) is true.

(B) $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x-y > 0))$

This says for all real x , y that if x is positive (or zero) and y is negative then $x-y$ is positive. This is true because if x is positive and y is negative, subtracting y from x will increase x and thus will be positive. i.e. for $x=1$, $y=-2$; $x-y=(1-(-2))=3>0$.

Therefore, (B) is true.

(C) $\exists x \exists y ((x \leq 0) \wedge (y \leq 0) \wedge (x-y > 0))$

The statement means there are some real numbers x and y such that both x and y are negative (or zero) and the difference $x-y > 0$. This condition will be satisfied when y is more negative than x . i.e. $x=-1$ and $y=-2$; $(x-y)=(-1-(-2))=-1>0$, So (C) is also true.

(D) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$

This condition means that for all x and y , x and y can't be zero if and only if their product xy isn't zero. This condition is always true because in real numbers, the product of two non-zero numbers is never zero. So, (D) is also true.

So option D is the correct answer.

Q42 Text Solution:

(A) $\forall n \exists m (n^2 < m)$:

This quantifier implies that for all integers n , there is an integer m that is greater than n^2 , it is false because the square of any number is greater than not less than.

(B) $\exists n \forall m (n < m^2)$

This quantifier implies there is an integer n such that for all integers m , n is less than m^2 . This is true if we pick n to be one of the negative integers.

(C) $\exists n \forall m (nm=m)$:

This quantifier implies that there's an integer n for which nm equals m for every integer m . This is true if we consider $n=1$ because $1 * m=m$ for all integer m .

(D) $\exists n \exists m (n^2+m^2=6)$

There exist integers n and m such that n^2+m^2 is



equal to 6. This is not true because there are no integers n and m such that their squares add up to 6.

(E) $\exists n \exists m (n+m=4 \wedge n-m=1)$

There exist integers n and m such that $n+m$ is equal to 4 and $n-m$ is equal to 1. This is false when $n=2$ and $m=1$.

So option B is the correct answer.

Q43 Text Solution:

The properties of the relation R are defined as $\langle x, y \rangle \in R \iff \langle u, v \rangle \iff xv=yu$

(A) Reflexive:

For a relation to be reflexive, (a, a) must be in the relation for every element a in the set. In this case, we have $x=u$ and $y=v$. So, (u, u) and (v, v) must be in the relation.

Since N is the set of natural numbers, for any natural number u or v , (u, u) and (v, v) are in the relation. Therefore, the relation is reflexive.

(B) Symmetric:

For a relation to be symmetric, if (a, b) is in the relation, then (b, a) must also be in the relation for every pair (a, b) . In this case, (x, y) is in the relation if and only if (y, x) is in the relation as $xv=yu$ implies $yu=xv$. Therefore, the relation is symmetric.

(C) Transitive:

For a relation to be transitive, if (a, b) and (b, c) are in the relation, then (a, c) must also be in the relation for every triplet (a, b, c) . In this case, if $xv=yu$ and $yw=zu$, then multiplying these equations gives $xw=zu$, which implies (x, w) is in the relation. Therefore, the relation is transitive.

(D) Asymmetric:

For a relation to be asymmetric, if (a, b) is in the relation, then (b, a) must not be in the relation for any pair (a, b) . In this case, since (x, y) is in the relation if and only if (y, x) is also in the relation, the relation cannot be asymmetric.

So option D is the correct answer.

Q44 Text Solution:

DMA (Direct Memory Access):

- DMA is a feature that allows peripherals to transfer data directly to and from the memory

without the intervention of the CPU.

- Instead of the CPU being involved in each data transfer between peripherals (like storage devices, network interfaces, etc) and memory, DMA enables the peripherals to communicate directly with the memory.
- This improves overall system efficiency by offloading data transfer tasks from the CPU.

IOP (Input-Output Processor):

- An IOP is a processor specifically designed to handle input and output operations. It serves as a dedicated unit for managing communication between peripherals and the rest of the system, including the memory.
- The IOP takes care of coordinating data transfers, handling interrupts, and managing other aspects of input and output it helps to offload these tasks from the main CPU, allowing it to focus on more complex computations.

So, both DMA and IOP are interfaces or features that facilitate the direct transfer of data between peripherals and the memory unit without requiring constant involvement of the main CPU. They contribute to improving the efficiency and performance of the overall system by streamlining data transfer operations.

Both serial and parallel interfaces can be used for data communication in computer systems, they are typically not used for direct communication with main memory in modern architectures.

Instead, high-speed serial interfaces like Peripheral Component Interconnect (PCI) are commonly employed for connecting main memory modules (RAM) directly to the CPU and other components on the motherboard.

So option C is the correct answer.

Q45 Text Solution:

Option A, Operating systems (OS) use virtual memory as a memory management strategy. This enables a computer to use secondary memory, such as a hard disk or solid-state drive (SSD), to temporarily boost the capacity of its main



memory, or RAM. Virtual memory controls the storage and retrieval of data from the hard drive through hardware and software. Without having to pay more for more RAM, virtual memory enables users to perform more on their PCs.

Option B, Cache, which is temporary memory storage, facilitates quicker and more effective data retrieval. It is the quickest memory in a computer and is usually found directly in the processor or main random access memory (RAM) of the motherboard.

Option C, RAM is a component of the main memory, which is also referred to as read-write memory. The motherboard has random access memory (RAM), which is used to temporarily store computer data. RAM helps with both reading and writing, as its name suggests. RAM is a volatile memory, meaning that once the computer switches off, the memory is gone. It is only present while the machine is turning on.

Option D, A solid-state storage device is called a solid-state drive (SSD). It offers permanent data storage. To offer a computer with relatively quick, reliable, direct-attached storage, an SSD is frequently used as secondary storage, not main memory. Although it lacks the disk-shaped component seen in a hard disk drive (HDD) or floppy disk, it is frequently referred to as a solid-state device or semiconductor storage device.

So option B is the correct answer.

Q46 Text Solution:

A tautology is a statement that is always true.

$$A: P \rightarrow Q = \neg P \vee Q$$

P	Q	$P \rightarrow Q$	$(P \rightarrow Q) = (\neg P \vee Q)$
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

So statement A is a tautology.

$$B: ((P \rightarrow Q) \rightarrow Q) \rightarrow (P \vee Q)$$

P	Q	$P \rightarrow Q$	$((P \rightarrow Q) \rightarrow Q)$	$(P \vee Q)$	$((P \rightarrow Q) \rightarrow Q) \rightarrow (P \vee Q)$
T	T	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	F	T	F	F	T

So statement B is a tautology.

$$C: ((P \vee \neg P) \rightarrow Q) \rightarrow ((P \vee \neg P) \rightarrow R)$$

P	Q	R	$(P \vee \neg P)$	$((P \vee \neg P) \rightarrow Q)$	$((P \vee \neg P) \rightarrow R)$	$((P \vee \neg P) \rightarrow Q) \rightarrow ((P \vee \neg P) \rightarrow R)$
T	T	F	T	T	F	F
T	F	F	T	F	F	T
T	F	T	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

C statement is not a tautology.

$$D: (Q \rightarrow (P \wedge \neg P)) \rightarrow (R \rightarrow (P \wedge \neg P))$$

P	Q	R	$(P \wedge \neg P)$	$(Q \rightarrow (P \wedge \neg P))$	$(R \rightarrow (P \wedge \neg P))$	$(Q \rightarrow (P \wedge \neg P)) \rightarrow (R \rightarrow (P \wedge \neg P))$
T	T	F	F	F	F	T
T	T	T	F	F	T	T
T	F	F	F	T	F	F
T	F	T	F	T	T	T
F	T	T	F	F	F	T
F	T	F	F	F	T	T
F	F	T	F	T	F	F
F	F	F	F	T	T	T



F	F	F	T	F	T	T	T
---	---	---	---	---	---	---	---

So statement D is also not a tautology.

Therefore option B is the correct answer, as statements A and B is a tautology.

Q47 Text Solution:

By looking at the lines that connect each point in question to the viewpoint (C), we can decide if one point obscures another. In particular, we must determine if the points P1, P2, and P3 that are provided are collinear that is, on the same line formed with viewpoint C. When viewed from the viewpoint, the point closer to the viewpoint will obscure the one farther away if the points are collinear.

here P1(1,2,0), P2(3,6,20) and P3(2,4,6) and a viewpoint C(0,0,-10)

Vector CP1: $(1-0, 2-0, 0-(-10))=(1,2,10)$; Vector CP2 $(3-0, 6-0, 20-(-10))=(3,6,30)$; Vector CP3: $(2-0, 4-0, 6-(-10))=(2,4,16)$

When all components of two vectors are multiplied by the same scalar value or one of the vectors is a scalar multiple of the other, the two vectors are said to be collinear. Therefore.

CP2=3CP1 since $(3,6,30)=3(1,2,10)$

CP3 is not any multiple of CP1, or of CP2 as $(2,4,16)$ is not any multiple of $(1,2,10)$ or $(3,6,30)$

So, from point C

(A) P1 obscures P2, as P2 lies on the line segment from C to P1.

(C) P3 does not obscure P1, as they do not lie on the same line segment.

(D) P2 does not obscure P3, as they do not lie on the same line segment.

So, the correct answer is: (A), (C) and (D) Only

Q48 Text Solution:

Statement 1: In C++, a constructor is a special function that is invoked automatically upon the creation of a class object. So statement A is correct.

The constructor name should be the same as the class name, enclosed in parenthesis ():

```
class MyClass { // The class
public:         // Access specifier
    MyClass() { // Constructor
```

```
    cout << "Hello World!";
}
};
int main() {
    MyClass myObj; // Create an object of
    MyClass (this will call the constructor)
    return 0;
}
```

Statement 2: Functions that don't return a value are known as constructors. Constructors, destructors, and void functions make up the family of functions that don't return a value. So statement B is correct.

Statement 3: When inheritance occurs, the derived class inherits all of the parent class's members (fields, methods, etc.), but it is not allowed to inherit the base class's constructor since constructors are not considered class members. It is only permitted to call the constructor of the base class; the derived class is not permitted to inherit constructors. So statement C is correct.

```
class Base
{
public:
    Base(char* name)
    {
        // ...
    }
};
class Derived : Base
{
public:
    Derived() : Base("hello") // calling the
    constructor of base class
    {
        // ...
    }
};
```

Statement 4: We cannot declare a virtual constructor. We should specify the exact type of the object at compile time so that the compiler can allocate memory for that specific type. So statement D is not correct. Therefore B is the correct answer.

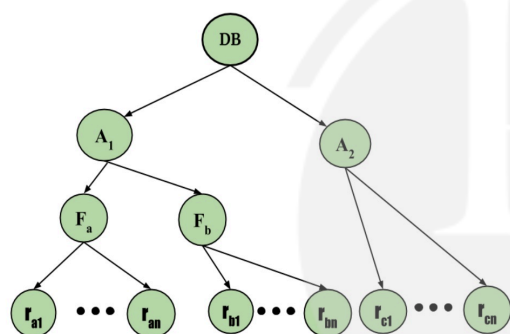


Q49 Text Solution:

Statement A - Granularity is the size of data items in a database.

It is the maximum size of a data item that can be locked. Multiple granularity refers to the hierarchical partitioning of the database into blocks that are trackable and able to be locked, depending on what has to be locked and how. A tree is a graphic representation of such a hierarchy.

Take the tree, for instance, which has four levels of nodes. The database as a whole is represented at the highest level. It is followed by nodes of type area; these are the areas that make up the database. Files are the children nodes in the area. Finally, each file has child nodes called records.



All of a node's descendants are implicitly locked in the same lock mode when a transaction locks a node, whether in shared or exclusive mode.

Statement B: Two operations in a schedule are said to be in conflict if they belong to the same transaction.

Two operations in a schedule are said to conflict if they access the same item X; and at least one of the operations is a write_item(X). For example, in schedule S1, the operations r1(X) and w2(X) conflict, as do the operations r2(X) and w1(X), and the operations w1(X) and w2(X). However, the operations r1(X) and r2(X) do not conflict, since they are both read operations; the operations w2(X) and w1(Y) do not conflict because they operate on distinct data items X and Y. So Statement B is correct.

Statement C : Two schedulers are said to be conflict equivalent if the order of any two

conflicting operations is the same in both schedules.

In a database system, a schedule is a sequence of operations (such as read and write operations) performed by transactions in the system. Two schedules are view-equivalent if they produce the same set of results when executed against the same database state.

A schedule S1 is said to be view-equivalent to a schedule S2 if and only if:

The order of any two conflicting operations in S1 is the same as the order of those operations in S2. A conflicting operation is an operation that accesses the same data item as another operation and at least one of the operations is a write operation.

So Statement C is correct.

Statement D: Write operations that are performed without performing the write operation are known as Blind Writes.

A blind write refers to a write operation on a database that does not first acquire locks on the data it is modifying. In a typical database system, before writing data, exclusive locks would be acquired to prevent other transactions from reading or writing the same data at the same time.

So statement D is incorrect.

Therefore option B is the correct answer.

Q50 Text Solution:

Statement A : The first record in each block of the data file is called the anchor record of that block not actor record. So statement A is not correct.

Statement B: Dense index has index entries for every search key value in the data file. It contains an entry for every data record. A dense index does not require the data file to be a sorted file. Instead, it can be built on any file organisation (typically, a heap file). So statement B is correct.

Statement C: In B+ tree data pointers are present only at the leaf nodes, the leaf nodes must necessarily store all the key values along with their corresponding data pointers to the disk file



block, to access them. Moreover, the leaf nodes are linked to providing ordered access to the records. The leaf nodes, therefore form the first level of the index, with the internal nodes forming the other levels of a multilevel index. So searching in B+ trees is much easier and faster than B-trees.

So statement C is not correct.

Statement D: Extendible hashing uses a directory to access its buckets. This directory is an array with 2^d entries kept in the main memory where each entry stores a bucket address (pointer to a bucket). The variable d is called the global depth of the directory.

So statement D is also not correct.

Therefore option B is the correct answer.

Q51 Text Solution:

Statement A: LOAD and STORE instructions are used to access remote memory.

Explanation: To load from memory, use the `ldr` instruction mnemonic followed by the target register and the memory address:

`ldr < register >, < memory address >.`

To store to memory, use the `str` instruction mnemonic followed by the source register and the memory address:

`str < register >, < memory address >`

e.g `LDR Rt, [Rn, immediate]`

`LDR Rt, [Rn, Rm]`

`STR Rt, [Rn, immediate]`

`STR Rt, [Rn, Rm]`

The instruction format for the LDR and Store Register (STR) instructions can add either an immediate or register value to the R_n value to calculate the memory address.

So statement A is correct.

Statement B and C: Explanation

In NUMA (NonUniform Memory Access) multiprocessors. To scale to more than 100 CPUs, they share a single address space across all the CPUs, but unlike UMA (uniform memory access) machines local access is faster than remote access.

So statement B is correct and C is not correct.

Statement D: When the access time to remote memory is hidden, the system is called NC - NUMA.

Explanation: All UMA programs run without change on NUMA machines, but their performance may be worse. When the access time to the remote machine is not hidden (by caching) the system is called NC-NUMA.

So Statement D is not correct.

Statement E: cache-coherent non-uniform-memory-access machines (CC-NUMA) and cache-only memory architectures (COMA). They both have distributed main memory and use directory-based cache coherence.

So statement E is correct.

Therefore D is the correct option.

Q52 Text Solution:

The allocation methods define how the files are stored in the disk blocks. There are three main disk space or file allocation methods.

1. Contiguous Allocation - In contiguous allocation, each file occupies a set of contiguous blocks on the disk.

2. Linked Allocation - In linked allocation, each file is a linked list of disk blocks. The allocated disk blocks may be scattered anywhere on the disk. The directory entry has a pointer to the first and the last blocks of the file.

3. Indexed Allocation-

- Indexed allocation supports sequential as well as direct access. For direct access, to access the i th block, the i th entry in the index block is taken. So statement A is correct.
- In indexed allocation all pointers are brought together into one block called the index block. Each file has an index block, which is an array of disk-block addresses. The i th entry in the index block points to the i th block of the file. So the entire block is available for data. Therefore statement B is correct.
- The limitation of indexed allocation is a waste of space. When there is a file occupying only two blocks, in linked allocation, space for only two pointers is wasted, in indexed allocation,



one whole block is wasted. So statement C is not correct.

There is no external fragmentation because any disk block from any position can be used. Contiguous allocation suffers from external fragmentation. As files are created and deleted, free disk blocks may not be present contiguously. But there may be free disk blocks here and there. The indexed allocation solves the external fragmentation, So statement D is correct. Therefore option D is the correct answer.

Q53 Text Solution:

Statement A: A process always check state of currently executing process to enter critical schema.

A process does not always need to check the state of the currently executing process to enter the critical section. It just needs to ensure mutual exclusion i.e. if another process is executing in its critical section, then no other process can enter the critical section. So statement A is incorrect.

Statement B: Spinlocks use busy waiting.

Spinlock is a synchronization mechanism where multiple processes access shared resources without interfering with each other. It is used in operating systems to protect shared resources from single access by multiple threads or processes. Spinlocks use a busy-wait method, where a thread continuously selects a lock until it becomes available. So statement B is correct.

Statement C: Critical region is a part of the program, where shared memory is kept.

A concurrent program's crucial region, sometimes referred to as its critical section, is the area where access to shared resources needs to be tightly controlled. This is required to ensure that errors or inconsistent program output are not caused by concurrent access to shared resources. So statement C is correct.

Statement D: Printer daemon, continuously checks to see if there are any file to be printed.

Printer and other output devices often respond to interrupts or other signals generated when a new print job arrives. This depends on the

implementation of the software. So statement D is incorrect.

Therefore option C is the correct answer.

Q54 Text Solution:

Three address code is a intermediate code which can be easily converted to machine code. It uses at most three addresses and one operator to represent an expression and the value computed at each instruction is stored in temporary variable generated by compiler. The compiler decides the order of operation given by three address code.

Option A, Quadruple – It is a structure that consists of 4 fields namely op, arg1, arg2 and result. op denotes the operator and arg1 and arg2 denotes the two operands and the result is used to store the result of the expression.

Option B, Triples – This representation doesn't make use of an extra temporary variable to represent a single operation instead when a reference to another triple's value is needed, a pointer to that triple is used. So, it consists of only three fields namely op, arg1, and arg2.

Option C, Pattern - In compiler design, it specifies a set of rules that a scanner follows to create a token. Tokens are a sequence of characters represented as a unit that cannot be further broken down.

Option D, Indirect Triples – This representation makes use of a pointer to the listing of all references to computations which is made separately and stored. It's similar in utility as compared to quadruple representation but requires less space than it. Temporaries are implicit and easier to rearrange code.

So (A), (B), and D are correct.

Q55 Text Solution:

Statement A: Moore and Mealy machines are finite state machines with output capabilities.

In the theory of computation, a Moore machine is a finite-state machine whose current output values are determined only by its current state.

This is in contrast to a Mealy machine, whose output values are determined both by its current



state and by the values of its inputs. So statement A is correct.

Statement B: Any given Moore machine has an equivalent Mealy machine.

Every Moore machine is equivalent to the Mealy machine with the same state, transitions and output function. So statement B is correct.

Statement C: Any given Mealy machine has an equivalent Moore machine.

Not every Mealy machine can be converted to an equivalent Moore machine. Some can be converted only to an almost equivalent Moore machine, with outputs shifted in time. This is due to the way that state labels are paired with transition labels to form the input/output pairs. So statement C is correct.

Statement D: Moore machine is not a finite state machine.

Moore and Mealy machines are both types of finite-state machines, they are equally expressive: either type can be used to parse a regular language. So statement D is not correct. Therefore option B is the correct answer.

Q56 Text Solution:

Given $P(x): x=x^2$

(A) $P(0): 0=0^2=0$ is true. So statement A is correct.

(B) $P(1): 1=1^2=1$ is true. So statement B is correct.

(C) $P(2): 2 \neq 2^2$ is false. So statement C is not correct.

(D) $\exists x P(x)$: that is there exist x such that $x=x^2$.

From statements (A) and (B) we can say that for $x=0$ and $x=1$ this is true. So statement D is correct.

(E) $\forall x P(x)$: that is for all x , $x=x^2$ which is not satisfied from statement (C) where $x=2$. So statement E is not correct.

Therefore (A), (B), and (D) are true. So option C is the correct answer.

Q57 Text Solution:

(A) $\forall n (n^2 \geq 0)$: This quantifier implies that for all integers n , n^2 is greater than equal to 0. It is true because the square of any integer number will be greater than or equal to 0.

(B) $\exists n (n^2 = 2)$: This quantifier implies that there exists an integer n whose square is equal to 2. It is false because $1^2=1$ and $2^2=4$. So statement B is not correct.

(C) $\forall n (n^2 \geq n)$: This quantifier implies that for all integers n , n^2 is greater than equal to n . It is true. e.g. $n=2$, $n^2=4 > 2$.

So statement C is correct.

(D) $\exists n (n^2 < 0)$: This quantifier implies that there exists an integer n whose square is less than 0. It is false because the square of an integer can't be a negative number. So statement D is false.

Therefore option D is the correct answer.

Q58 Text Solution:

An undirected graph is a tree if it has the following properties.

1. There is no cycle - We can either use BFS or DFS. For every visited vertex ' v ', if there is an adjacent ' u ' such that u is already visited and u is not the parent of v , then there is a cycle in the graph. If we don't find such an adjacent for any vertex, we say that there is no cycle.

2. The graph is connected - Since the graph is undirected, we can start BFS or DFS from any vertex and check if all vertices are reachable or not. If all vertices are reachable, then the graph is connected, otherwise not.

Therefore, if a graph is connected and has $n - 1$ edges exactly then the graph is a tree.

From (A) - vertex $n=6$ and edges = 5, So A forms a tree.

From (B) - vertex $n=6$ and edges = 5, So B forms a tree.

From (C) - vertex $n=6$ and edges = 6, So C does not form a tree.

From (D) - vertex $n=6$ and edges = 4, So D does not form a tree.

Therefore option A is the correct answer.

Q59 Text Solution:

A node using the carrier-sense multiple access (CSMA) medium access control (MAC) protocol confirms that there isn't any other traffic before sending data over a shared transmission medium,



like an electrical bus or an electromagnetic spectrum band. Before starting a transmission under CSMA, a transmitter uses a carrier-sense mechanism to ascertain whether another transmission is already in progress.

Variations of CSMA use different algorithms to determine when to initiate transmission onto the shared medium. A key distinguishing feature of these algorithms is how persistent they are in initiating transmission.

(A) 1-persistent - When the transmitting node is ready to transmit, it senses the transmission medium for idle or busy. If idle, then it transmits immediately. If busy, then it senses the transmission medium continuously until it becomes idle. In case of a collision, the sender waits for a random period of time and attempts the same procedure again.

(B) 2-persistent - It is not an example of CSMA channel sensing methods.

(C) p-persistent- When the transmitting node is ready to transmit data, it senses the transmission medium for idle or busy. If idle, then it transmits immediately. If busy, then it senses the transmission medium continuously until it becomes idle, then transmits with probability p . If the node does not transmit (the probability of this event is $1-p$), it waits for a random period of time before re-attempting the same procedure, using the same probability p . If the transmission medium is not busy, it transmits again with the same probability p .

(D) o-persistent - When the transmission medium goes idle, nodes wait for their time slot in accordance with their assigned transmission order. The node assigned to transmit first transmits immediately. The node assigned to transmit the second waits one-time slot (but by that time the first node has already started transmitting). Nodes monitor the medium for transmissions from other nodes and update their assigned order with each detected transmission. So Option B is the correct answer.

Q60 Text Solution:

The Spiral Model in software development is a risk-driven model that involves four phases: planning, risk analysis, engineering, and evaluation. This model is ideal for large, complex, and innovative projects when frequent releases are necessary.

(A) If requirements are easily understandable and defined, the Waterfall model is used. It is frequently employed for lengthy, large-scale projects with tight deadlines where we need minimal margin for mistakes and a high degree of stakeholder faith in the final product.

(B) Do we change requirements quite often? - Software development can be done incrementally and iteratively using the Spiral Model, which enables flexibility and adaptability in response to unanticipated events or changing requirements.

(C) Can we define requirements early in the cycle? The waterfall model is suitable for projects when the requirements are clear and steady and reduces modifications as the project progresses.

(D) Requirements are indicating a complex to be built - The spiral model is suitable for complex projects as the risk analysis phase allows for thorough identification and mitigation of any potential issues in a complex project. The spiral approach is beneficial for projects with moderate to high risk when requirements are complicated and ambiguous.

So option C is the correct answer.

Q61 Text Solution:

User Datagram Protocol (UDP) is a Transport Layer protocol. UDP is a part of the Internet Protocol suite, referred to as UDP/IP suite. UDP does not provide-

- congestion control - how much data can be sent at a time.
- flow control - to make sure a fast sender can not overwhelm a slow receiver.
- retransmission- Where lost packets are detected.



UDP permits packets to be dropped instead of processing delayed packets so there is no retransmission. There is no error checking in UDP, so it also saves bandwidth. Unlike TCP, it is an unreliable and connectionless protocol. There is no need to establish a connection before data transfer. So it does not provide flow and congestion control.

- Segment Transmission- The UDP enables process-to-process communication. Each UDP segment is handled individually from others as each segment takes a different path to reach the destination. The UDP segments can be lost or delivered out of order to reach the destination as there is no connection setup between the sender and the receiver.

In process-to-process communication, the client generates the processes that need services while the server generates the processes that provide services. The queues are available for both processes, i.e., two queues for each process. The first queue is the incoming queue that receives the messages, and the second one is the outgoing queue that sends the messages. The queue functions when the process is running. If the process is terminated then the queue will also get destroyed.

So option C is the correct answer.

Q62 Text Solution:

(A) OSPF is based on distance-vector routing protocol.- This statement is FALSE.

Explanation: Using its own shortest path first (SPF) algorithm, open shortest path first (OSPF) is a link-state routing protocol that determines the optimal path between the source and the destination router. In contrast to distance-vector routing protocols, which interchange the routing table over time, link-state routing protocols rely solely on the concept of triggered updates—that is, updates are only sent in response to changes noticed in the learned routing table.

(B) Both link-state and distance-vector routing are based on the least cost goal- This statement

is TRUE.

Distance Vector Routing –

It is a dynamic routing algorithm in which each router computes a distance between itself and each possible destination i.e. its immediate neighbors.

The router shares its knowledge about the whole network with its neighbors and accordingly updates the table based on its neighbors.

The sharing of information with the neighbors takes place at regular intervals.

Link State Routing –

It is a dynamic routing algorithm in which each router shares knowledge of its neighbors with every other router in the network.

A router sends its information about its neighbors only to all the routers through flooding.

Information sharing takes place only whenever there is a change.

(C) BGP4 is based on the path-vector algorithm - This statement is TRUE.

A path-vector routing protocol is a network routing protocol that maintains the path information that gets updated dynamically. Updates that have looped through the network and returned to the same node are easily detected and discarded. Border Gateway Protocol (BGP) is a standardized exterior gateway protocol designed to exchange routing and reachability information among autonomous systems (AS) on the Internet. BGP is classified as a path-vector routing protocol, and it makes routing decisions based on paths, network policies, or rule sets configured by a network administrator.

(D) The three-node instability can be avoided using a split-horizon combined with poison reverse.- This statement is FALSE

Counting to infinity is a drawback of distance-vector routing. While cost increases (bad news) propagate slowly, cost decreases (positive news) propagate quickly. Distance-vector routing takes some time for all other routers to become aware of a broken connection (cost becomes infinity),



which is necessary for a routing protocol to function properly.

- Split-horizon route advertisement is a method of preventing routing loops in distance-vector routing protocols by prohibiting a router from advertising a route back onto the interface from which it was learned.
- With poison reverse, route advertisements that would be suppressed by split horizon are instead advertised with a distance of infinity.

With the help of split horizon and poison reverse, the two-node instability can be prevented.

However, stability cannot be assured if instability occurs between three nodes. The instability between three nodes is referred to as three-node instability

(E) RIP is based on the link state algorithm - This statement is FALSE.

Routing Information Protocol (RIP) is a dynamic routing protocol that uses hop count as a routing metric to find the best path between the source and the destination network.

So option A is the correct answer.

Q63 Text Solution:

(A) NRZ is a bipolar scheme in which the positive voltage defined bit is 0 (zero).- This statement is FALSE.

Explanation: A non-return-to-zero (NRZ) line code is a binary code in which ones are represented by one significant condition, usually a positive voltage, while zeros are represented by some other significant condition, usually a negative voltage, with no other neutral or rest condition.

(B) NRZ-L and NRZ-I both have an average signal rate of $N / 2$.- This statement is TRUE.

Explanation: NRZ-L (Nonreturn to Zero-Level) is the code used to generate or interpret digital data by terminals and other devices. If a different code is to be used for transmission, it is generated from an NRZ-L signal by the transmission system.

As with NRZ-L, NRZI maintains a constant voltage pulse for the duration of a bit of time. The data themselves are encoded as the presence or absence of a signal transition at the beginning of the bit time. A transition (low to high or high to low) at the beginning of a bit time denotes a binary 1 for that bit time; no transition indicates a binary 0. In both signal drops (but does not return to zero) in the middle of the bit period. The average signal rate in both cases is $N / 2$.

(C) The idea of RZ and NRZ-L are combined into the Manchester scheme. - This statement is TRUE.

Explanation: Manchester encoding is a synchronous clock encoding technique used by the physical layer of the Open System Interconnection [OSI] to encode the clock and data of a synchronous bit stream. Differential Manchester, combines the idea of RZ and NRZ-I. There is always a transition in the middle of the bit, but the bit values are determined at the beginning of the bit. if the next bit is zero there is a transition if the next bit is 1 there is none.

(D) NRZ-L and NRZ-I both have DC component problems.- This statement is TRUE

Explanation: When the voltage level remains constant for long periods of time, there is an increase in the low frequencies of the signal. Most channels are band pass and may not support the low frequencies. This is known as DC problem. both NRZ-L and NRZ-I maintain a constant voltage level for the entire bit period. So both have DC component problems.

(E) The minimum bandwidth of Manchester and differential Manchester is 3 times that of NRZ. - This statement is FALSE.

Explanation: The Manchester and differential Manchester Encoding consumes twice the bandwidth of the original signal not 3 times. So option C is the correct answer.

Q64 Text Solution:

Syntactic analysis or parsing or syntax analysis is the third phase of NLP. The purpose of this phase



is to draw exact meaning, or dictionary meaning from the text.

(A) Top-Down Parsing- A parse tree is a tree that defines how the grammar was utilized to construct the sentence. Using the top-down approach, the parser attempts to create a parse tree from the root node down to the leaves.

(B) Bottom-Up Parsing - It begins with the words of input and attempts to create trees from the words up, by applying grammar rules one at a time. The parse is successful if it builds a tree rooted in the start symbol S that includes all of the input.

(C) Dependency parsing - It shows grammatical links between words by constructing a tree structure in which each word in the sentence is dependent on another.

(D) Statistical machine translation- Statistical machine translation (SMT) is not a parsing technique. It is a type of machine translation (MT) that uses statistical models to translate text from one language to another.

(E) Earley parsing - The Earley parser is an algorithm for parsing strings that belong to a given context-free language.

So option C is the correct answer.

Q65 Text Solution:

(A) Alpha Beta pruning can eliminate subtrees with certainty when the value of a node exceeds both the alpha and beta bounds. This is correct. If a node's value falls outside the range defined by alpha and beta, it can be pruned.

(B) The primary purpose of Alpha-Beta pruning is to save computation time by searching fewer nodes in the same tree. This is correct. Alpha-Beta pruning reduces the number of nodes that need to be examined in the search tree, thereby speeding up the computation.

(C) Alpha Beta pruning guarantees the optimal solution in all cases by exploring the entire game tree. This statement is incorrect. While alpha-beta pruning does find the optimal solution, it does not do so by exploring the whole tree. The whole point of alpha-beta pruning is to avoid having to

explore the entire tree.

(D) Alpha and Beta bounds are initialized to negative and positive infinity respectively at the root node. This one is correct. At the start of search, alpha (the value of the best option we have found so far for max) is initialized to negative infinity, and beta (the value of the best option we have found so far for min) is initialized to positive infinity.

So option C (Statements A, B and D) is the correct answer.

Q66 Text Solution:

In the context of Set Theory:

- $\emptyset$ denotes the empty set, also written as $\{\}$.
- $\cap$ denotes intersection, which means it gives the common elements in both sets.
- $\cup$ denotes a union, which gives all unique elements from both sets.
- $-$ denotes set difference, which means it gives elements of the first set that are not in the second set.

(A) $\phi \cap \{\phi\}$ is the intersection of an empty set with a set containing an empty set. This results in an empty set. So, (A) matches with (I).

(B) $\{\phi\} \cap \{\phi\}$ is the intersection of a set containing an empty set with another set containing an empty set. This results in a set containing an empty set. So, (B) matches with (II).
(C) $\{\phi, \{\phi\}\} - \phi$, Set difference gives elements of the first set that are not in the second set. This matches with $\{\phi, \{\phi\}\}$ (IV).

(D) $\phi \cup \{\{\phi\}\}$ is the union of an empty set with a set containing an empty set. This results in a set containing only the set that contains an empty set. So, (D) matches with (III).

So option D ((A)-(I), (B)-(II), (C)-(IV), (D)-(III)) is the correct answer.

Q67 Text Solution:

a) SZA: "Skip if Zero Accumulator" instruction. the accumulator (AC) register in assembly language is used to hold intermediate arithmetic and logic results. SZA says the next instruction would be skipped by the program if the AC is



zero.

b) SKI: "Skip if Input flag is on". A value that serves as a signal for a function or process is known as a flag in computer programming. If a particular input flag was raised or turned "on," this command will skip the subsequent one.

c) SNA: "Skip if Accumulator is Negative," says if there is a negative value in the accumulator, this instruction, SNA, would skip the subsequent operation.

d) ISZ: "Increment and Skip if Zero" means the designated memory location (M) would be incremented by one, and then if the result is zero, the next instruction would be skipped.

So option B, (a)-(IV), (b)-(III), (c)-(II), (d)-(I) is the correct answer.

Q68 Text Solution:

a) Bresenham's line algorithm is a line drawing algorithm. It is commonly used to draw line primitives in a bitmap image (e.g. on a computer screen), as it uses only integer addition, subtraction, and bit shifting.

b) In computer graphics, the Cohen–Sutherland algorithm is an algorithm used for line clipping. The algorithm divides a two-dimensional space into 9 regions and then efficiently determines the lines and portions of lines that are visible in the central region of interest (the viewport).

c) The Sutherland–Hodgman algorithm is used for clipping polygons. It works by extending each line of the convex clip polygon in turn and selecting only vertices from the subject polygon that are on the visible side.

d) Z buffer is also known as Depth buffer, it is the simplest and most widely used algorithm, it is an image space approach used for hidden surface detection.

So (a)-(II), (b)-(III), (c)-(IV), (d)-(I) option C is the correct answer.

Q69 Text Solution:

a) BCNF iff - BCNF: A relation is in BCNF if and only if it meets two conditions:

- Every non-trivial functional dependency (FD) $X \rightarrow Y$ (where X determines Y) in the relation, X

must be a superkey. Non-trivial means Y is not a subset of X.

- There are no non-trivial multi-valued dependencies (MVDs) in the relation.
- Implication by Candidate Keys: If every FD in the relation can be derived from the candidate keys, then the relation is in BCNF. So IV matches for BCNF.
- b) 5NF iff - The relation between 5NF and Join Dependency is that a relation is in 5NF if it is in 4NF and does not have any join dependencies.
- A join dependency is a constraint on the set of legal relations over a database scheme. A table T is subject to a join dependency if T can always be recreated by joining multiple tables each having a subset of the attributes of T .
- If one of the tables in the join has all the attributes of the table T , the join dependency is called trivial.
- A table is said to be in the 5NF if and only if every non-trivial join dependency in that table is implied by the candidate keys. So I matches for 5NF.
- c) 1NF iff- A relation is in first normal form if and only if each attribute contains only atomic, or indivisible, values, hence the statement "all underlying domains contain scalar values only". So, II matches for 1NF.
- d) 4NF iff - A multivalued dependency exists when there are at least three attributes (like X, Y and Z) in a relation and for a value of X there is a well defined set of values of Y and a well defined set of values of Z. However, the set of values of Y is independent of set Z and vice versa.
- A table is in 4NF if and only if, for every one of its non-trivial multivalued dependencies $X \twoheadrightarrow Y$, X is a superkey that is, X is either a candidate key or a superset of a key. So it matches III.

Therefore correct matches are (a)-(IV), (b)-(I), (c)-(II), (d)-(III), option B.

Q70 Text Solution:



- a) LRU replaces the line in the cache that has been in the cache the longest with no reference to it. It works on the idea that the more recently used blocks are more likely to be referenced again. So it matches with II.
- b) Demand cleaning is the process in which a page is written to secondary memory only when it has been selected for replacement. So it matches with I.
- c) An operating system component called a long-term scheduler chooses which processes should be allowed into the system and when. It functions at a high level and is utilized in batch-processing systems. Allocating processor time and memory to processes according to their priorities and demands is the responsibility of the long-term scheduler. In addition, it chooses the sequence in which operations are carried out and controls the execution of lengthy activities, like batch jobs or background tasks. So it matches with IV.
- d) Medium-term scheduling- On a system that does not use virtual memory, memory management also becomes an issue that must be addressed by the medium-term scheduler. This means that the swapping-in decision must consider the memory requirements of the swapped-out process. So medium-term scheduling ensure that processes are not using more memory than they are supposed to. i.e. it takes the decision to add to the number of processes that are partially or fully in main memory. So it matches III.
- Therefore (a)-(II), (b)-(I), (c)-(IV), (d)-(III), Option B is the correct answer.

Q71 Text Solution:

a) Syntax tree- Semantic Analysis is the third phase of the Compiler. Semantic Analysis makes sure that declarations and statements of the program are semantically correct. Semantic Analyzer uses a syntax tree and symbol table to check whether the given program is semantically consistent with the language definition. It gathers type information and stores it in either a

syntax tree or symbol table. This type of information is subsequently used by the compiler during intermediate code generation. So a Syntax tree corresponds to a Semantic analyzer.

b) Intermediate representation- A code generator is a compiler that translates the intermediate representation of the source program into the target program. In other words, a code generator translates an abstract syntax tree into machine-dependent executable code. So intermediate representation corresponds to code generator.

c) Token stream-Syntax analysis or parsing is the second phase of a compiler design process. Lexical analyzer which is the first phase of compiler design can identify tokens with the help of regular expressions and pattern rules. But a lexical analyzer cannot check the syntax of a given sentence due to the limitations of the regular expressions. Regular expressions cannot check balancing tokens, such as parenthesis. Therefore, the syntax analyzer phase takes the input from a lexical analyzer in the form of token streams. The parser analyzes the source code (token stream) against the production rules to detect any errors in the code. The output of this phase is a parse tree.

d) Character stream - Lexical Analysis is the very first phase in the compiler design. A Lexer (Lexical analyzer) takes the modified source code which is written in the form of sentences. It converts a sequence of characters into a sequence of tokens.

Therefore option B is the correct answer.

Q72 Text Solution:

- a) Since $P \rightarrow Q = \neg P \vee Q$
 $P \wedge (P \rightarrow Q)$
 $P \wedge (\neg P \vee Q)$ (by implication law)
 $(P \wedge \neg P) \vee (P \wedge Q)$ (by distributive law)
 b)



$\neg(P \vee Q) \rightarrow (P \wedge Q)$
 $\neg(\neg(P \vee Q)) \vee (P \wedge Q)$
(by implication law)
 $(P \vee Q) \vee (P \wedge Q)$
(by negation property)
 $(P \vee (Q \vee (P \wedge Q)))$ *(by associative law)*
 $(P \vee ((Q \vee P) \wedge (Q \vee Q)))$
(by distributive law)
 $(P \vee ((Q \vee P) \wedge Q))$
(by Idempotent law)
 $(P \vee (P \vee Q) \wedge Q)$ *(by commutative law)*
 $(P \vee (P \vee (Q \wedge Q)))$ *(by associative law)*
 $(P \vee (P \vee Q))$ *(by idempotent law)*
 $(P \vee P) \vee Q$ *(by associative property)*
 $P \vee Q$ *(by iempotent law)*

c) $P \rightarrow Q$
 $\neg P \vee Q$ *(by implication law)*
 $(\neg P) \vee Q$ *(by associative property)*

d)
 $P \vee (Q \wedge R)$
 $(P \vee Q) \wedge (P \vee R)$
 $(P \wedge (P \vee R)) \vee (Q \wedge (P \vee R))$
(by distributive property)
 $(P \wedge P) \vee (P \wedge R) \vee (Q \wedge P) \vee (Q \wedge R)$
(by idempotent and distributive law)
 $(P \wedge P) \vee (P \wedge R) \vee (P \wedge Q) \vee (Q \wedge R)$
(by commutative property)
 $(P \wedge P) \vee (P \wedge Q) \vee (P \wedge R) \vee (Q \wedge R)$
(by associative property)

So option B is the correct answer.

Q73 Text Solution:

- a) Greedy Best First Search- Greedy Best-First Search is not a complete algorithm, meaning it may not always find a solution if one exists. This can happen if the algorithm gets stuck in a cycle or if the search space is too complex. It corresponds to "Incomplete even if the search space is finite".
- b) A* algorithm can be computationally expensive in certain scenarios when the search space is extensive and the number of possible

paths is large. The algorithm may consume significant memory and processing resources. Its space complexity is $O(b^d)$ where d is the depth of the solution (the shortest path) and b is the branching factor (the average number of successors per state), as it stores all generated nodes in memory. So it corresponds to "Computation and space complexity is two light".

c) Recursive best-first search- When all nodes are explored, the best node is explored but the second best is preserved in memory. If the best node is explored and all children are found to be worse, the search process backtracks to the second-best node (which is now best in the list including all the children as well as it). At the same point of time, the best child is kept as second best node now. At any given point in time, this search method only explores one node and remembers one more. So the space complexity is $O(d)$ where d = depth of the deepest optimal solution.

d) SMA* or Simplified Memory Bounded A* is a shortest path algorithm based on the A* algorithm. The main advantage of SMA* is that it uses a bounded memory. It is complete if the allowed memory is high enough to store the shallowest solution. It is optimal if the allowed memory is high enough to store the shallowest optimal solution, otherwise, it will return the best solution that fits in the allowed memory. It avoids repeated states as long as the memory bound allows it. So it corresponds to III.

Therefore Option A is the correct answer.

Q74 Text Solution:

- a. Monoalphabetic cipher - Monoalphabetic uses only one alphabet. An alphabetic substitution is a substitution cipher where the letters of the alphabet are replaced by others according to a 1-1 correspondence (a plain letter always corresponds to the same cipher letter).
- b. DES- Round-specific data derived from the cipher key called a round key. A key schedule is an algorithm that calculates all the round keys from the key. DES has a key schedule in which



the 56-bit key is divided into two 28-bit halves; each half is thereafter treated separately. In successive rounds, both halves are rotated left by one or two bits (specified for each round), and then 48 round key bits are selected by Permuted Choice 2 (PC-2) – 24 bits from the left half and 24 from the right. The rotations have the effect that a different set of bits is used in each round key; each bit is used in approximately 14 out of the 16 round keys. A key schedule is an algorithm that calculates all the round keys from the key.

c. Stream Cipher- A linear feedback shift register (LFSR) is a shift register whose input bit is a linear function of its previous state. The most commonly used linear function of single bits is exclusive-or (XOR). Thus, an LFSR is most often a shift register whose input bit is driven by the XOR of some bits of the overall shift register value. LFSRs have long been used as pseudo-random number generators for use in stream ciphers, due to the ease of construction from simple electromechanical or electronic circuits, long periods, and very uniformly distributed output streams. However, an LFSR is a linear system, leading to fairly easy cryptanalysis.

d. Polyalphabetic cipher- In polyalphabetic substitution, the relationship between a character in the plaintext and the characters in the ciphertext is one-to-many.

Therefore option A is the correct answer.

Q75 Text Solution:

All these results are based on the size of a search tree – the number of nodes in a tree, with (average) branching factor b and depth d . This space is dominated by the size of the queue (the list of partial paths).

a) Hill Climbing - The best-case scenario occurs when the local maximum is found in the first step. In this case, the space complexity is $O(1)$.

b) BFS- BFS adds nodes to the end of the queue. So, by the time the algorithm has arrived at the right-most leaf node, it will have pushed all the nodes in the search tree on its queue. So worst case space complexity of BFS is $O(b^d)$.

c) $A^* - A^*$ has the same worst-case complexity as BFS. So Worst case space complexity of A^* is $O(b^m)$.

d) DFS- In the worst case, suppose the Goal lies at a leaf node of the search tree. In this case, at each iteration of the search, DFS will add b partial paths to the queue. At most, it can do this d times, until it reaches the bottom (leaf node) of the search tree. At this point, the queue is of size bd . If the Goal is not found at this point, DFS unwinds (the stack is popped), perhaps all the back to the first level. So bd is the maximum space used. So the Worst case space complexity of DFS is $O(bd)$.

So option A is the correct answer.

Q76 Text Solution:

Fermat's little theorem states that if p is a prime number, then for any integer a , the number $a^p - a$ is an integer multiple of p . In the notation of modular arithmetic, this is expressed as $a^p \equiv a \pmod{p}$

If a is not divisible by p , that is, if a is coprime to p , then Fermat's little theorem is equivalent to the statement that $a^{p-1} - 1$ is an integer multiple of p , or in symbols:

$$a^{p-1} \equiv 1 \pmod{p}$$

Option A) Remainder of 49^{16} when divided by 17: here $a=49$ and $p=17$

$$\text{so } 49^{17-1} \equiv 1 \pmod{17}$$

$$49^{16} \equiv 1 \pmod{17}$$

The remainder of 49^{16} when divided by 17 is 1.

b) Remainder of 2^{446} when divided by 9

$$2^6 = 64 \% 9 = 1$$

$$2^{10} = 1024 \% 9 = 7$$

$$2^{20} = 2^{10} \cdot 2^{10} = 7 \cdot 7 = 49 \% 9 = 4$$

$$2^{40} = 2^{20} \cdot 2^{20} = 4 \cdot 4 = 16 \% 9 = 7$$

$$2^{80} = 2^{40} \cdot 2^{40} = 7 \cdot 7 = 49 \% 9 = 4$$

$$(2^{80})^5 = 2^{400} = 4^5 = 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 = 1024 \% 9 = 7$$

$$2^{446} = 2^{400} \cdot 2^{40} \cdot 2^6 = 7 \cdot 7 \cdot 1 = 49 \% 9 = 4$$

So Remainder of 2^{446} when divided by 9 is 4

c) Remainder of 155^{17} when divided by 17

$$155 \% 17 = 2$$

$$(155)^2 = 2 \cdot 2 = 4 \% 17 = 4$$

$$(155)^4 = 4 \cdot 4 = 16 \% 17 = 16$$



$$(155)^8 = 16.16 = 256\%17 = 1$$

$$(155)^{16} = 1.1 = 1\%17 = 1$$

$$(155)^{16} \cdot 155 = 1.2 = 2\%17 = 2$$

So Remainder of 155^{17} when divided by 17 is 2

d) Last digit of number 7^{45}

$$\text{Since } 7^1 = 7$$

$$7^2 = 49$$

$$7^3 = 343$$

$$7^4 = 2401$$

$$7^5 = 16807$$

As we can see the last digits 7,9,3,1 are repetitive after every power. So the last digits of the number 7^{45} ($45 \bmod 4 = 1$) will be **7**.

After arranging them in ascending order we will get a, c, b, d (according to the values 1,2,4,7).

So option C is the correct answer.

Q77 Text Solution:

Explanation:

In computer organization, an instruction cycle, also known as a fetch-decode-execute cycle, is the basic operation performed by a central processing unit (CPU) to execute an instruction. The instruction cycle consists of several steps, each of which performs a specific function in the execution of the instruction.

Fetch: In the fetch cycle, the CPU retrieves the instruction from memory. The instruction is typically stored at the address specified by the program counter (PC). The PC is then incremented to point to the next instruction in memory.

Decode: In the decode cycle, the CPU interprets the instruction and determines what operation needs to be performed. This involves identifying the opcode and any operands that are needed to execute the instruction.

Calculate the effective address. If the instruction has an operand, it's not usually the actual data but rather its address that is specified. During this phase, the address for the operands is computed. The calculated effective address will be used to fetch the operand in the next step.

Fetch operands: In some CPUs, the operands needed for an instruction are fetched during a separate cycle before the execute cycle. This is called the fetch operands cycle.

Execute: In the execute cycle, the CPU performs the operation specified by the instruction. This may involve reading or writing data from or to memory, performing arithmetic or logic operations on data, or manipulating the control flow of the program.

So option B is the correct answer.

Q78 Text Solution:

```
#include <stdio.h>
int main() {
    int arr[]={0,1,2,3,4};
    int i=1, *ptr;
    ptr =arr+2 ;
    printf ("% d", ptr [i]);
    printf ("%d", ptr [i+1]);
    printf ("%d", ptr [-i]);
    printf ("%d", ptr [-i+1]);
    return 0;
}
```

Output - 3412

Explanation:

- $\text{ptr} = \text{arr} + 2$; sets ptr to point to the third element of the array arr, which is 2.
- $\text{ptr}[i]$ is equivalent to $*(\text{ptr} + i)$ i.e. value at index 3 which is **3**.
- so $\text{ptr}[i+1] = 4$
- since $\text{ptr} = \text{arr} + 2$, which points value at index 2, so $\text{ptr}[-i]$ will give value at index 1 which is **1**.
- $\text{ptr}[-i+1]$ will point to $\text{ptr}[0]$ which is **2**. Therefore if we want to arrange the print statements in increasing order it would be Option B (c,d,a,b).

Q79 Text Solution:

An analysis model created using object-oriented analysis is transformed by object-oriented design into a design model that works as a plan for software creation. The steps for the analysis and design of an object-oriented system are-



Draw a Use case model and activity diagram:
The first step in the analysis and design of an object-oriented system is to recognize the actors interlinked with the system. After that, create the use case and draw the use case diagram. The activity Diagram demonstrates the dynamic nature of a system by creating the flow of control from activity.

Draw the interaction diagram :

An interaction diagram shows a collaboration, comprising a bunch of articles and their relationship, including the messages that might be dispatched among them.

Draw the class diagram: The class diagram is responsible for showing the relationship between the classes.

Design of State chart diagrams :

A state chart is utilized to show the state space of a given class, the occasion that influences progress starting with one state then onto the next, and the activity that outcome from a state change.

Draw component and development diagram :

These diagrams address the static execution perspective on a system they are identified with class diagrams in that a segment ordinarily guides to at least one class, interface, or coordinated effort.

So option D is the correct answer.

Q80 Text Solution:

Interrupt: An interrupt is an event that alters the sequence of instructions executed by a processor in corresponding to the electrical signal generated by the HW circuit both inside & outside the CPU.

Following are the uses and priority of the interrupts in Linux kernel-

(a) user mode programs- These are applications that are currently running on our computer. They have the lowest degree of system privileges and don't communicate directly with other programs or the system hardware. They ask the kernel for system services and resources.

(b) kernel system service routines- These procedures provide system services, which help the user mode programs. Since they deal directly with hardware and system structures, they function at a higher privilege level than user programs, but they are not as prioritized as interrupt handlers.

(c) bottom half interrupt handlers - The Job of the Bottom half is used to run leftover (deferred) work by the top halves. When this piece of code is being executed interrupt is Enabled and Scheduler is Disabled. The bottom half then performs whatever other work is required, such as awakening processes, starting up another I/O operation, and so on. This setup permits the top half to service a new interrupt while the bottom half is still working. Every serious interrupt handler is split this way.

(d) top half interrupt handlers- Top halves execute as soon as the CPU receives the interrupt. In the Top Halves, Context Interrupt and Scheduler are disabled. This part of the code only contains Critical Code. The execution time of this code should be as short as possible because at this time Interrupt is disabled. When a top-half handler is called, it will handle the hardware operations and then add the bottom-half handler(s) to the interrupt queue because it is the highest priority.

So the correct answer is option B.

Q81 Text Solution:

A compiler is a software program that converts the high-level source code written in a programming language into low-level machine code that can be executed by the computer hardware.

The typical phases of a compiler are:

- Lexical Analysis: The first phase of a compiler is lexical analysis, also known as scanning. This phase reads the source code and breaks it into a stream of tokens, which are the basic units of the programming language.
- Syntax Analysis: The second phase of a compiler is syntax analysis, also known as



parsing. This phase takes the stream of tokens generated by the lexical analysis phase and checks whether they conform to the grammar of the programming language.

- **Semantic Analysis:** The third phase of a compiler is semantic analysis. This phase checks whether the code is semantically correct, i.e., whether it conforms to the language's type system and other semantic rules.
- **Intermediate Code Generation:** The fourth phase of a compiler is intermediate code generation. This phase generates an intermediate representation of the source code that can be easily translated into machine code.
- **Optimization:** The fifth phase of a compiler is optimization. This phase applies various optimization techniques to the intermediate code to improve the performance of the generated machine code.
- **Code Generation:** The final phase of a compiler is code generation. This phase takes the optimized intermediate code and generates the actual machine code that can be executed by the target hardware.

So option C is the correct answer.

Q82 Text Solution:

The prototype model is a prototype of the end product is first developed, tested, and refined as per customer feedback repeatedly till a final acceptable prototype is achieved.

Steps of Prototyping Model

Step 1: Requirement Gathering and Analysis: This is the initial step in designing a prototype model.

In this phase, users are asked about what they expect or what they want from the system.

Step 2: Quick Design: This is the second step in the Prototyping Model. This model covers the basic design of the requirement through which a quick overview can be easily described.

Step 3: Implement: This step implements an actual prototype from the knowledge gained from prototype design.

Step 4: Customer Evaluation: This step describes the preliminary testing where the investigation of the performance model occurs, as the customer will tell the strengths and weaknesses of the design, which was sent to the developer.

Step 5: Design: If any feedback is given by the user, then improving the client's response to feedback and suggestions, the final system is designed.

So option C is the correct answer.

Q83 Text Solution:

- In Genetic algorithms, a chromosome (also sometimes called a genotype) is a set of parameters that define a proposed solution to the problem that the evolutionary algorithm is trying to solve.
- Encoding determines the possible values for each gene (It is one element position in the chromosome.) in a chromosome. Selecting the correct encoding is one of the challenges in GA and it is one of the most common mistakes.

In permutation encoding, every chromosome is a string of numbers that represent a position in a sequence. Permutation encoding is useful for ordering problems.

Example of Problem: Travelling salesman problem (TSP)

The problem: There are cities and given distances between them. The traveling salesman has to visit all of them, but he does not want to travel more than necessary. Find a sequence of cities with a minimal traveled distance.

Encoding: Chromosome describes the order of cities, in which the salesman will visit them.

Gray coding is a binary numeral system where two successive values differ in only one bit. In the context of genetic algorithms. Gray code takes a binary sequence and shuffles it to form new sequences.

In binary encoding, every chromosome is a string of bits - 0 or 1. Binary encoding represents



a gene in terms of 0 or 1, so a chromosome is represented by a string of binary bits.

Example of Problem: Knapsack problem

The problem: There are things with a given value and size. The knapsack has given capacity. Select things to maximize the value of things in the knapsack, but do not extend the knapsack capacity.

Encoding: Each bit says, whether the corresponding thing is in knapsack. i.e. 0 = exclude the item, and 1 = include the item. We can then sum the included items' values to find how well that chromosome performs.

Real-value encoding represents a gene with numeric values, strings, or symbols. The solution is expressed in its natural state. This encoding is used when potential solutions contain continuous values that cannot be encoded easily with binary encoding.

Example of Problem: Finding weights for a neural network

The problem: A neural network is given with a defined architecture. Find weights between neurons in the neural network to get the desired output from the network.

Encoding: Real values in chromosomes represent weights in the neural network.

- So encoding schemes where fitness depends on order only - Permutation
- encoding schemes where fitness depends on value and order - Binary
- encoding schemes where fitness depends on value only - Real value

We start with ordering i.e. permutation encoding. Then we will try to shuffle to form new sequences using Gray coding. From Gray coding we will get binary encoding, then we will use real-value encoding because real-value encoding is used when potential solutions contain continuous values that cannot be encoded easily with binary encoding.

So option C is the correct answer.

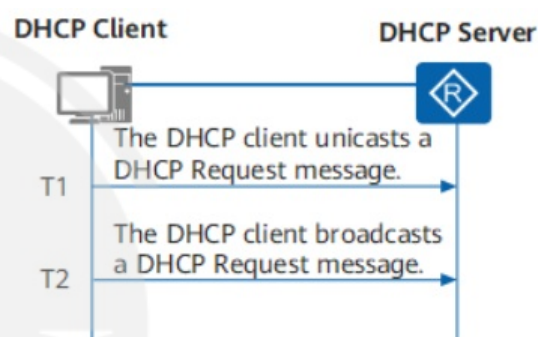
Q84 Text Solution:

IP addresses that are dynamically allocated by a DHCP server have leases. However, a DHCP client may request a specific lease term by adding information to a DHCP Discover message.

Following are the steps for a DHCP Client to renew its IP lease with a DHCP server:

c) DHCP client checks the local lease timer and initiates renewal when the lease reaches 50% of its validity period, DHCP clients manage a lease timer that monitors the duration for which the allocated IP address is valid.

a) DHCP client sends a DHCPREQUEST message- After checking, the DHCP client unicasts a DHCP Request message to the DHCP server to request lease renewal.



b) DHCP server acknowledges the renewal with a DHCPACK message- If the server determines that the request is valid and the IP address can be renewed, the server sends back a DHCPACK message to the client.

(d) DHCP server updates its lease database- After successfully renewing the client's lease, the DHCP server updates its lease database to reflect this change.

So option D is the correct answer.

Q85 Text Solution:

K-Means Clustering is an unsupervised learning algorithm that is used to solve the clustering problems in machine learning or data science. The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).



Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassigning each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.

So option A is the correct answer.

Q86 Text Solution:

Statement (I): If H is a nonempty finite subset of a group G and $ab \in H \forall a, b \in H$ then H is also a group

This statement is correct under the condition that H also contains the identity element of G and that for each element in H , its inverse is also in H . These are key requirements to being a subgroup and therefore a group on its own.

Statement (II): There is no homomorphism exist from $(Z, +)$ to $(Q, +)$, where Z is the set of integers and Q is the set of rational numbers.

This statement is false. Let (G, o) & (G', o') be 2 groups, a mapping "f" from a group (G, o) to a group (G', o') is said to be a homomorphism if –
 $f(aob) = f(a) o' f(b) \forall a, b \in G$

A homomorphism does exist from $(Z, +)$ to $(Q, +)$. One such example could be the identity function $f(x)=x$ where both addition operations remain preserved.

So option C is the correct answer.

Q87 Text Solution:

Statement I: The friend function and the member functions of a friend class directly access the private and protected data is true because A friend class can access private and protected members of other classes in which it is declared as a friend. It is sometimes useful to allow a particular class to access private and protected members of other classes.

```
class base {
private:
```

```
    int private_variable;

protected:
    int protected_variable;

public:
    base()
    {
        private_variable = 10;
        protected_variable = 99;
    }

    // friend class declaration
    friend class F;
};

// Here, class F is declared as a
// friend inside class base. Therefore,
// F is a friend of class base. Class F
// can access the private and protected members
// of
// class base.
class F {
public:
    void display(base& t)
    {
        cout << "The value of Private Variable = "
        << t.private_variable << endl;
        cout << "The value of Protected Variable = "
        << t.protected_variable;
    }
};

// Driver code
int main()
{
    base g;
    F fri;
    fri.display(g);
    return 0;
}

The value of Private Variable = 10
The value of Protected Variable = 99

Statement II: "The friend function can access the private data through the member functions
```



of the base class" is also true because Similar to a friend class, a friend function can be granted special access to private and protected members of a class in C++. They are the non-member functions that can access and manipulate the private and protected members of the class for they are declared as friends.

```
class base {
private:
    int private_variable;

protected:
    int protected_variable;

public:
    base()
    {
        private_variable = 10;
        protected_variable = 99;
    }

    // friend function declaration which is
    member function of base class
    friend void friendFunction(base& obj);
};

// friend function definition
void friendFunction(base& obj)
{
    cout << "Private Variable: " <<
obj.private_variable
    << endl;
    cout << "Protected Variable: " <<
obj.protected_variable;
}

// driver code
int main()
{
    base object1;
    friendFunction(object1);

    return 0;
}
```

Private Variable: 10

Protected Variable: 99

So option A, Both Statement I and Statement II are correct.

Q88 Text Solution:

Statement (I): A thread is a dispatchable unit of work that does not execute sequentially and is not interruptible is false because a process, a thread refers to a single sequential activity being executed. these activities are also known as thread of execution or thread control.

Threads run in parallel improving the application performance. Each such thread has its own CPU state and stack, but they share the address space of the process and the environment. Priority can be assigned to the threads just like the process, and the highest priority thread is scheduled first. Each thread has its own Thread Control Block (TCB). Like the process, a context switch occurs for the thread, and register contents are saved in (TCB). As threads share the same address space and resources, synchronization is also required for the various activities of the thread. So these are also interruptible.

Statement (II): It is not possible to alter the behaviour of a thread by altering its context when the thread is suspended -Threads also have states like ready, executing, blocked, etc. A thread's register values, program counter, and other execution-related data make up its context. The context of a suspended thread can be preserved, changed, and then restored when the thread starts up again. This permits modifications to the thread's behavior.

So option B is the correct answer.

Q89 Text Solution:

Reuse Oriented Model (ROM), also called reuse-oriented development (ROD), is a process where software is redesigned by building a series of prototypes, or models. Each system is developed based on the previous one and follows a set of constant rules.

The reuse model has 4 fundamental steps which are followed :



1. To identify components of the old system that are most suitable for reuse.
2. To understand all system components.
3. Modify old system components to achieve new requirements.
4. To integrate all of the modified parts into the new system

So statement I is correct and II is incorrect.

Option C is the correct answer.

Q90 Text Solution:

Datagram switching is done at the network layer of the communication system.

1. In datagram networks, each data packet or datagram is routed independently from the source to the destination even if they belong to the same message. The network treats the packet as if it exists alone.
2. Since the datagrams are treated as independent units, no dedicated path is fixed for data transfer. Each datagram is routed by the intermediate routers using dynamically changing routing tables. So two successive packets from the source may follow completely separate routes to reach destination. So there is no need of routers to hold state information about connections.

So statement I is incorrect.

Virtual – circuit network is a category of packet switching network, where a virtual path is established between the source and the destination systems for data communication to occur.

Phases of Virtual - Circuit Transmission

There are three phases of transmission by virtual circuits, set up, data transfer and teardown.

1. Set up Phase – In this phase, a virtual circuit or a route is established from the source to the destination through number of switches. The source and destination use global addresses using which the switches make routing table entries.
2. Data Transfer – Once the virtual circuit is set up, all packets follow the route established

during the set up phase adhering to the routing tables.

3. Teardown Phase – When data transfer is complete, the source sends a teardown request. The destination responds using a teardown confirmation. The switches flush their routing table entries, thus relinquishing the circuit. Each router maintains a separate entry in the routing table for each virtual circuit that it is a part of.

So statement II is correct.

Q91. Text Solution:

Cost of Food X per gram = Rs 12

Cost of Food Y per gram = Rs 20

The cost function of total food can be expressed as the sum of the cost of Food X and the cost of Food Y:

$\text{Cost}(x, y) = \text{Cost of Food X} + \text{Cost of Food Y}$

Given that x is the quantity (in grams) of Food X, and y is the quantity (in grams) of Food Y, the total cost function becomes:

$\text{Cost}(x, y) = (12 * x) + (20 * y)$

$Z = 12x + 20y$

Q92. Text Solution:

To formulate a Linear Programming Problem (LPP), we need to consider the following constraints:

1. Vitamin D Constraint: The total amount of Vitamin D obtained from both Food X and Food Y should meet or exceed the minimum daily requirement, which is 100 units. Mathematically, this constraint can be expressed as:
 $6x + 8y \geq 100$
2. Vitamin E Constraint: The total amount of Vitamin E obtained from both Food X and Food Y should meet or exceed the minimum daily requirement, which is 120 units. Mathematically, this constraint can be expressed as:
 $7x + 12y \geq 120$
3. Non-negativity Constraint: The quantities x and y representing the amount of Food X



and Food Y purchased cannot be negative.
Mathematically, this constraint can be expressed as:
 $x \geq 0, y \geq 0$

So option D is the correct answer.

Q93. Text Solution:

The objective function (cost function):

$$Z = 12x + 20y$$

Constraints:

$$\text{Vitamin D constraint: } 6x + 8y \geq 100$$

$$\text{Vitamin E constraint: } 7x + 12y \geq 120$$

$$\text{Non-negativity constraint: } x \geq 0, y \geq 0$$

We need to find the values of x and y that minimize the cost function while satisfying these constraints.

putting $x=0$ in $6x+8y \geq 100$

$$8y = 100$$

$$y = 100/8 = 25/2 = 12.5$$

when $y=0$ in $6x+8y \geq 100$

$$x = 100/6 = 50/3 = 16.6$$

putting $x=0$ in $7x+12y \geq 120$

$$12y = 120$$

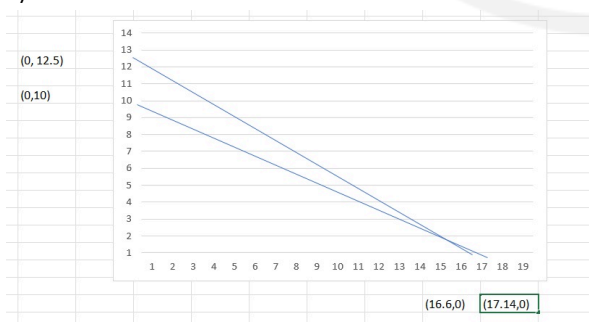
$$y = 10$$

When $y=0$

$$7x = 120$$

$$x = 120/7 = 17.14$$

By graphical method, we can see that the intersection points of these two lines are 15 and $5/4$



Since it minimizes the problem so feasible solutions lie on three points only (0,12.5), (16.6,0), and (17.14,0)

x	y	$z = 12x + 20y$
0	25/2	250
120/7	0	205.71
15	5/4	205

so option B is the correct answer.

Q94. Text Solution:

Objective function (cost function):

$$\text{Min } Z = 12x + 20y$$

Constraints:

$$\text{Vitamin D constraint: } 6x + 8y \geq 100$$

$$\text{Vitamin E constraint: } 7x + 12y \geq 120$$

$$\text{Non-negativity constraint: } x \geq 0, y \geq 0$$

The dual of this LPP can be obtained by converting it into a maximization problem. Let (u) and (v) be the dual variables associated with the constraints.

Maximize:

$$W = 100u + 120v$$

subject to

$$6u + 7v \leq 12$$

$$8u + 12v \leq 20$$

$$u, v \geq 0$$

So option A is the correct answer.

Q95. Text Solution:

Values of X and Y when cost of food is minimum

$$\text{is } X = 15 \text{ and } Y = \frac{5}{4}$$

$$Z = 12x + 20y$$

$$= 12 \cdot 15 + 20 \cdot \frac{5}{4}$$

$$= 180 + 25$$

$$= 205$$

So Option A is the correct answer.

Q96. Text Solution:

The SQL LEFT OUTER JOIN is the types of the outer join to combine the two tables. To get the left join output using SQL, it finds all the rows from the first table including the matching rows from the right table. However, if there is no match in the second table it returns a null value. So from the given two tables left outer join will be

S	Si	C	P	D
J	1	M	S1	CA
B	2	N	P1	AB
R	3	H	D1	DC
T	4	G	NULL	NULL

From the above table, we can see there are 4 tuples.



So option C is the correct answer.

Q97. Text Solution:

Here we need to find out Right outer join-Right outer join query result finds all the rows from the second table and includes the matching rows from the left table, if there is no match in both the tables it returns a null value.

So output will be Option B

S	Si	C	P	D
J	1	M	S1	CA
B	2	N	P1	AB
R	3	H	D1	DC
A	NULL	NULL	H1	MD

Q98. Text Solution:

The FULL OUTER JOIN keyword returns all records when there is a match in left (table1) or right (table2) table records with Null values of corresponding missing values.

S	Si	C	P	D
J	1	M	S1	CA
B	2	N	P1	AB
R	3	H	D1	DC
T	4	G	NULL	NULL
A	NULL	NULL	H1	MD

So option D is the correct answer.

Q99. Text Solution:

Right outer join query result finds all the rows from the second table and includes the matching rows from the left table, if there is no match in both the tables it returns a null value.

S	Si	C	P	D
J	1	M	S1	CA
B	2	N	P1	AB
R	3	H	D1	DC
A	NULL	NULL	H1	MD

So Option D will be the correct answer Since there are 4 tuples.

Q100. Text Solution:

The Cartesian product is a set formed from two or more given sets and contains all ordered pairs of elements such that the first element of the pair is from the first set the second is from the second set, and so on.

There are 4 tuples from the left outer join and 4 from the right outer join, So Number of tuples obtained by applying cartesian product over X and Y will be $4 \times 4 = 16$.

So option A is the correct answer.

